Fluid Mechanics ME-280

Based on Tobias Schneider's Lectures

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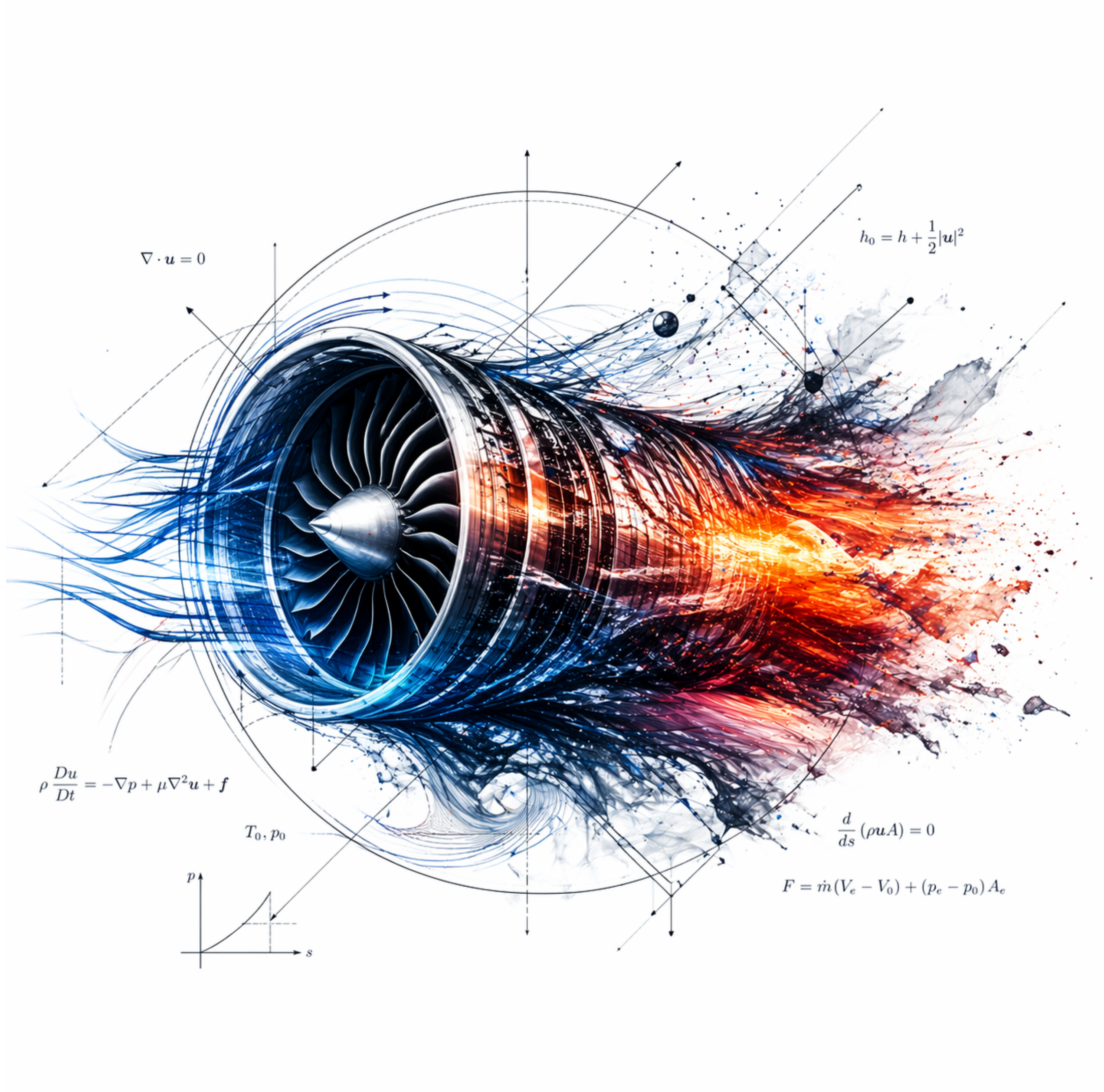

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https://github.com/Examera1005/Fluid-Mechanics-ME280

Contents

1 Week 1 - Introduction to Fluid Mechanics

1.1 Definition of a Fluid

Definition: Fluid Definition

A **fluid** is a substance that deforms continuously under the application of any shear stress, no matter how small. Alternatively, it can be defined at the molecular scale as a collection of molecules that cannot statically resist a shear stress.

- **Tangential Stress:** A stress acting parallel to the surface interface.
- **Threshold Behavior:** The magnitude of the shear stress does not matter in fluids. If a minimum non-zero threshold shear stress is required to initiate motion or deformation in an object, that object is categorized as a yield-stress material (e.g., Bingham plastic) and is **not** a true fluid.
- **Shear Resistance:** The internal structural resistance of a fluid to continuous shear deformation is dynamically determined by its **viscosity**.

1.1.1 Axiomatic Laws of Continuum Fluid Dynamics

Every macro-scale fluid dynamics phenomenon must strictly satisfy the following fundamental physical conservation laws:

1. **Newton's Second Law:** Conservation of linear momentum.
2. **Conservation of Mass:** The continuity principle.
3. **Conservation of Energy:** The First Law of Thermodynamics.
4. **The Second Law of Thermodynamics:** (Will probably not be dealt with in this course).
5. **Newton's Third Law:** Action-reaction balance across internal fluid interfaces.

1.2 Thermodynamic and Physical Properties of Fluids

1.2.1 Primary Properties

- **Density (ρ):** Mass per unit volume:

$$\rho = \frac{m}{V} \tag{1}$$

- **Specific Volume (v):** Volume per unit mass:

$$v = \frac{1}{\rho} \tag{2}$$

- **Specific Weight (γ):** Gravitational force per unit volume:

$$\gamma = \rho g \tag{3}$$

- **Specific Gravity (SG):** Dimensionless ratio relative to water at standard reference conditions:

$$\mathrm{SG} = \frac{\rho}{\rho_{\mathrm{H_2O}}} \tag{4}$$

1.2.2 State Dependencies

From classical thermodynamics, density is a function of the local state variables: $\rho = f(T, P)$. This holds **only when temperature and pressure are independent variables**.

Under standard baseline reference conditions ($P = 1$ atm, $T = 300$ K), typical pure fluid density values are:

$$\rho_{\mathrm{H_2O}} = 1000\,\mathrm{kg\,m^{-3}} \tag{5}$$

$$\rho_{\mathrm{Air}} = 1.22\,\mathrm{kg\,m^{-3}} \tag{6}$$

1.2.3 Ideal Gas Equation of State

For compressible gas phases operating far from their critical point, the density field is governed by:

$$\rho = \frac{P}{RT} \quad (\text{or } P = \rho RT) \tag{7}$$

Where P is absolute pressure, T is absolute temperature (K), and R (or R_s) is the specific gas constant.

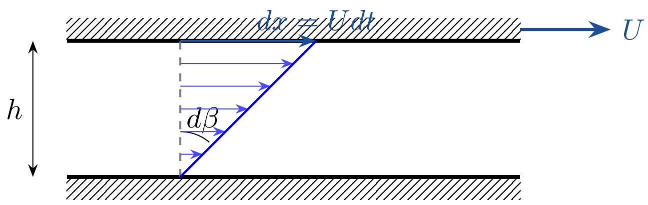
1.3 Kinematic Foundations of Viscosity

1.3.1 The No-Slip Condition

The fundamental kinematic boundary condition states that: At a solid physical boundary, a viscous fluid will exhibit zero velocity relative to that boundary.

1.3.2 Derivation of Shear Strain Rate

Consider a fluid layer of height h trapped between a stationary plate and a top plate translating at a velocity U. Over an infinitesimal time increment dt, the top surface translates by a distance $dx = Udt$:



Evaluating the angular deformation geometry for small angles ($d\beta \ll 1$):

$$\tan(d\beta) = \frac{dx}{h} \approx d\beta \tag{8}$$

Differentiating this angular deformation with respect to time yields the continuous strain rate:

$$\frac{d\beta}{dt} = \frac{1}{h}\frac{dx}{dt} = \frac{U}{h} = \dot{\gamma} \tag{9}$$

This kinematic property is defined as the **Shear Strain Rate**. For linear profiles, $\frac{U}{h} \approx \frac{du}{dy}$.

1.4 Rheological Classification & Constitutional Equations

1.4.1 Newtonian Fluids

A fluid is classified as **Newtonian** if the internal shear stress τ is directly proportional to the applied shear strain rate $\dot{\gamma} = \frac{du}{dy}$:

$$\tau \propto \frac{du}{dy} \implies \tau = \mu \frac{du}{dy} \tag{10}$$

Where μ is the **Dynamic Viscosity**, with SI dimensions $[\mu] = \mathrm{kg\,m^{-1}\,s^{-1}} = \mathrm{Pa\,s}$.

The physical analogue of a Newtonian fluid is a Hookean solid, with the distinction that solid stress depends on absolute displacement strain ($\sigma \propto \epsilon$), whereas fluid stress depends strictly on the strain *rate* ($\tau \propto \dot{\gamma}$).

1.4.2 Kinematic Viscosity

To decouple viscous diffusion from inertial fluid mass scales, the kinematic viscosity ν is defined as:

$$\nu = \frac{\mu}{\rho} \quad \left([\nu] = \mathrm{m^2\,s^{-1}}\right) \tag{11}$$

Alternative Viscosity Units

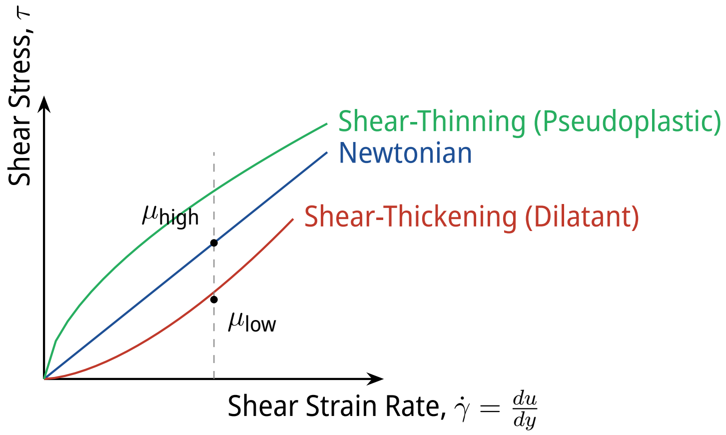
A common legacy CGS unit for dynamic viscosity is the poise (P), defined as $1\,\mathrm{g\,cm^{-1}\,s^{-1}}$. The centipoise ($\mathrm{cP} = 1 \times 10^{-3}\,\mathrm{Pa\,s}$) is frequently implemented since $\mu_{\mathrm{H_2O}} \approx 1\,\mathrm{cP}$ at room temperature.

Standard approximate reference orders of magnitude for dynamic viscosity (μ) to memorize:

Fluid Phase	Dynamic Viscosity μ [Pa s]
Air	1.8×10^{-5}
Water	1.1×10^{-3}
Engine Oil	0.4

1.4.3 Non-Newtonian Rheology

In complex dimensional configurations or multi-phase fluid flows, the stress-strain relation becomes non-linear. These systems are governed by effective apparent viscosities that deviate from Equation 10:



- **Shear-Thinning (Pseudoplastic):** Apparent viscosity decreases with increasing strain rate (represented by the lower fluid locus circle where the local slope $\frac{d\tau}{d\dot{\gamma}}$ flattens out).
- **Shear-Thickening (Dilatant):** Apparent viscosity increases continuously with increasing strain rate (the curve exhibits exponential-like upward growth).

1.5 Measure of Compressibility

To evaluate volume variations under uniform structural pressure changes ($V, P \to P(V)$), we define the **Bulk Modulus of Elasticity** (E_V):

$$E_V = -V \left(\frac{\partial P}{\partial V}\right)_T = \rho \left(\frac{\partial P}{\partial \rho}\right)_T \tag{12}$$

1.6 Introduction to Hydrostatics

1.6.1 Pressure at a Point

While mechanical stress is a second-order tensor ($\boldsymbol{\sigma} = \frac{\vec{F}}{A}$ holding a directional orientation), hydrostatic pressure is treated strictly as a **scalar field**.

This scalar behavior is proven analytically by isolating a differential physical control mass element (real or imaginary fluid element), and matching Newton's second law with bounding surface force distributions in the zero-velocity limit.

1.7 Advanced Problem Set 1: Multi-Layer Viscous Couette Flow

1.7.1 Problem Statement

Consider two immiscible, incompressible fluid layers bounded between a fixed bottom floor and a top plate translating at a constant horizontal velocity U. The lower layer has dynamic viscosity μ_2 and thickness $(1-c)E$, while the upper layer has dynamic viscosity $\mu_1 = \frac{1}{2}\mu_2$ [cite: 66] and thickness cE[cite: 76, 77].

Determine the exact multi-dimensional velocity vector $\vec{v}$ [cite: 69] and evaluate the velocity magnitude v exactly at the internal fluid-fluid interface boundary[cite: 76].

1.7.2 Analytical Derivation

Isolating an infinitesimal interface element, Newton's third law dictates that the internal shear stresses across the fluid-fluid boundary must balance exactly to avoid an unphysical infinite acceleration of the boundary mass ($\Sigma F_x = 0$)[cite: 75]:

$$\tau_1 = \tau_2 \implies \mu_1 \frac{du_1}{dy} = \mu_2 \frac{du_2}{dy} [cite: 75, 76] \tag{13}$$

Assuming fully developed, linear Couette velocity profiles within each discrete fluid zone, the matching strain rates evaluated at the interface yield[cite: 76]:

$$\tau_1 = \mu_1 \frac{v}{cE} \quad \text{and} \quad \tau_2 = \mu_2 \frac{U-v}{(1-c)E} [cite: 76] \tag{14}$$

Substituting the explicit viscosity ratio parameter $\mu_2 = 2\mu_1$ into the boundary stress balance equation[cite: 66, 77]:

$$\mu_1 \frac{v}{cE} = 2\mu_1 \frac{U-v}{(1-c)E} [cite: 76, 77] \tag{15}$$

Canceling out the common dynamic scale factors μ_1 and thickness base E yields[cite: 77]:

$$\frac{v}{c} = \frac{2(U-v)}{1-c} [cite: 77] \tag{16}$$

Expanding the linear algebraic cross-multiplication[cite: 78]:

$$v(1-c) = 2c(U-v) [cite: 78] \tag{17}$$

$$v(1-c) = 2Uc - 2vc [cite: 78, 79] \tag{18}$$

Grouping all internal interface velocity components v onto the left-hand side[cite: 79, 80]:

$$v(1-c) + 2vc = 2Uc [cite: 79] \tag{19}$$

$$v(1-c+2c) = 2Uc \implies v(1+c) = 2Uc [cite: 80, 83] \tag{20}$$

1.7.3 Asymptotic Evaluation

Solving explicitly for the interface velocity component v[cite: 83]:

Interface Boundary Velocity Solution

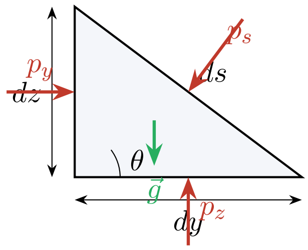
$$v = \frac{2Uc}{1+c}[cite:83] \tag{21}$$

Evaluating the mathematical correctness of the derived continuous constitutional equation under physical boundary limits:

- **Limit $c \to 0$:** The upper fluid domain layer vanishes completely. The interface velocity drops to zero ($v = 0$)[cite: 83], which satisfies the stationary lower wall boundary condition perfectly.
- **Limit $c \to 1$:** The lower fluid domain layer vanishes completely. The interface matches the upper translation plate limits ($v = U$)[cite: 83], satisfying the no-slip constraint at the moving ceiling.

2 Week 2 - Pressure at a Point (Isotropy & Scalarity)

Consider a microscopic, wedge-shaped fluid element suspended in an air-water system below a free surface, subject to a gravitational acceleration field $\vec{g}$:



By applying a localized static force balance ($\Sigma\vec{F} = \vec{0}$) on the element while explicitly neglecting shearing forces ($P = \frac{F}{A}$):

$$p_z \cdot dxdy - p_s \cdot dsdx\cos(\theta) - \rho g dV = \rho dV a_z \tag{22}$$

In the vertical z-direction, this force balance maps directly to:

$$p_z - p_s - \rho g\frac{dz}{2} = \rho a_z\frac{dz}{2} \tag{23}$$

Taking the continuous limit as the geometric dimensions vanish ($dx, dy, dz \to 0$):

$$p_z = p_s \tag{24}$$

Performing an identical scale analysis along the horizontal y-direction (where the localized component of gravitational weight is identically zero):

$$p_y = p_s \tag{25}$$

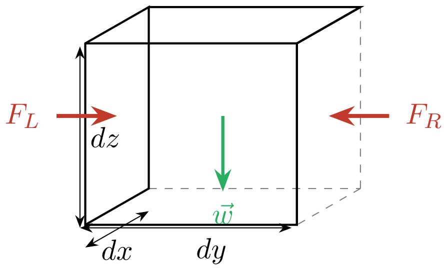
Isotropy of Hydrostatic Pressure

Because $p_y = p_z = p_s$ under the limit $dV \to 0$, pressure is identically uniform in all directions at a single spatial point. Consequently, pressure behaves strictly as a **scalar field**.

2.1 Governing Equation for the Pressure Field

2.1.1 Taylor Series Expansion on an Infinitesimal Fluid Volume

To derive the continuous differential spatial distribution of pressure, we examine an isolated differential fluid cube of volume $dV = dxdydz$ centered around a point O:



Using a first-order multivariate Taylor-series expansion, we define the localized pressures acting on the left (P_L) and right (P_R) faces bounding the y-axis:

$$P_R = P + \left.\frac{\partial P}{\partial y}\right|_0 \frac{dy}{2} + \mathcal{O}(dy^2) \tag{26}$$

$$P_L = P - \left.\frac{\partial P}{\partial y}\right|_0 \frac{dy}{2} + \mathcal{O}(dy^2) \tag{27}$$

Converting these localized scalar fields into their corresponding differential force vectors ($F = P \cdot A$) acting across the profile area $dA = dxdz$:

$$F_R = P_R\, dxdz = \left(P + \left.\frac{\partial P}{\partial y}\right|_0 \frac{dy}{2}\right) dxdz \tag{28}$$

$$F_L = P_L\, dxdz = \left(P - \left.\frac{\partial P}{\partial y}\right|_0 \frac{dy}{2}\right) dxdz \tag{29}$$

2.1.2 Multi-Dimensional Vector Formulation

Applying Newton's second law along the y-axis profile while strictly neglecting viscous shearing stresses:

$$-F_R + F_L = \rho\, dV a_y \implies -\left.\frac{\partial P}{\partial y}\right|_0 dxdydz = \rho\, dxdydz\, a_y \tag{30}$$

Evaluating the limit as $dV \to 0$ yields the continuous differential equation for the acceleration component:

$$-\left.\frac{\partial P}{\partial y}\right|_0 = \rho a_y \tag{31}$$

Extending this identical structural logic symmetrically across the orthogonal x and y spatial directions:

$$-\frac{\partial P}{\partial x} = \rho a_x, \quad -\frac{\partial P}{\partial y} = \rho a_y \tag{32}$$

Combining these spatial derivatives via the vector gradient operator ∇P:

$$-\nabla P = -\frac{\partial P}{\partial x}\hat{\imath} - \frac{\partial P}{\partial y}\hat{\jmath} - \frac{\partial P}{\partial z}\hat{k} \tag{33}$$

Incorporating the body force vector of fluid weight acting strictly down the vertical axis ($\vec{w} = -\rho\, dV g\hat{k}$), Newton's second law formulated per unit volume becomes:

General Hydrodynamic Equation of Motion

$$-\nabla P - \rho g\hat{k} = \rho\vec{a} \tag{34}$$

2.2 Hydrostatics: Fluids at Rest

2.2.1 The Hydrostatic Differential Equation

For a fluid configured completely at rest, the global acceleration vector is identically null ($\vec{a} = \vec{0}$). The general equation of motion reduces to:

$$-\nabla P - \rho g\hat{k} = \vec{0} \implies \frac{\partial P}{\partial x} = 0, \quad \frac{\partial P}{\partial y} = 0, \quad \frac{\partial P}{\partial z} = -\rho g \tag{35}$$

Expert Analysis: Density Field Dependencies

This shows mathematically that for a fluid at rest, the pressure field depends strictly on the vertical coordinate ($P = P(z)$). The global behavior of this field is governed by the structural equation of state for fluid density $\rho = \rho(P, T)$.

2.2.2 Incompressible Fluid Integration

For liquids or fluids modeling an constant density profile (ρ = cst), the differential equation integrates explicitly across an arbitrary vertical distance $\Delta z = z_2 - z_1$:

$$\int_{P_1}^{P_2} dP = \int_{z_1}^{z_2} -\rho g\, dz \implies \Delta P = -\rho g \Delta z \tag{36}$$

Defining $h = z_0 - z$ as the downward vertical depth measured relative to a baseline reference height z_0 at pressure P_0:

$$P - P_0 = \rho g h \tag{37}$$

2.2.3 Pressure Scale Conventions

Definition: Absolute vs. Gauge Pressure

Pressure measurements are categorized based on their underlying reference datum:

- **Absolute Pressure (P_{abs}):** Measured relative to a perfect physical vacuum ($P_{\text{vacuum}} = 0$).
- **Gauge Pressure (P_{gauge}):** Measured relative to the local ambient atmospheric pressure (P_{atm}).

The mathematical relation connecting the two domains is defined by:

$$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} \tag{38}$$

3 Week 3 - Hydrostatics: Variable Density & Manometry

Recall the fundamental equation of fluid statics: $\frac{dP}{dz} = -\rho g$. For liquids, ρ is constant. For gases, it varies.

3.1 Ideal Gas Law & Isothermal Atmosphere

Using the Ideal Gas equation $P = \rho RT \implies \rho = \frac{P}{RT}$.

Scientific Verification: Ideal Gas Law

The formulation $P = \rho RT$ used in engineering was consolidated by Émile Clapeyron in 1834, combining Boyle's, Charles's, and Avogadro's empirical laws into a single rigorous equation of state.

Substituting this into the hydrostatic equation for an isothermal condition ($T =$ cst):

$$\int_{P_0}^{P} \frac{dP}{P} = \int_0^z -\frac{g}{RT}\, dz$$

$$\ln\left(\frac{P}{P_0}\right) = -\frac{g}{RT} z \implies P(z) = P_0 \exp\left(-\frac{gz}{RT}\right)$$

3.2 Manometry (Measuring Pressure)

A manometer uses liquid columns to measure pressure differences. By stepping down and up the tube (pressure increases going down, decreases going up):

$$P_1 + \rho_{\text{fluid}} g z_1 - \rho_m g h = P_{\text{atm}}$$

Scientific Verification: The Manometer

The concept of using fluid columns to measure differential pressure was invented by Christiaan Huygens around 1661. It directly relies on Stevin's and Pascal's principles that pressure in a continuous fluid only depends on depth.

3.3 Forces on Submerged Surfaces

3.3.1 Flat Surfaces & Center of Pressure

Consider a flat surface of area A inclined at an angle θ. The pressure increases linearly with depth, creating a pressure prism.

Resultant Force and Location

- Magnitude: $F_R = P_c \cdot A = (\rho g h_c) A$

- Location (Center of Pressure): $y_{cp} = y_c + \frac{I_{xc}}{y_c A}$

Where I_{xc} is the area moment of inertia. The Parallel Axis Theorem (Huygens, 1673) proves that $y_{cp} > y_c$, meaning the force always acts below the centroid due to the increasing pressure with depth.

3.3.2 Curved Submerged Surfaces: Tank Sidewall Example

For a curved surface, it is easier to calculate the horizontal and vertical components separately. Consider a tank with a curved quarter-circle sidewall of radius R and width w.

Vertical Component (F_V): Equals the weight of the fluid volume directly above the curved surface.

$$F_V = \int_V \rho g \, dV = \rho g \left(\underbrace{h \cdot R \cdot w}_{\text{Rectangle}} + \underbrace{\frac{\pi R^2}{4} w}_{\text{Quarter circle}} \right)$$

Horizontal Component (F_H): Equals the force acting on the vertical projection of the curved surface (a rectangle of height R and width w).

$$F_H = \int_{A_v} P \, dA = \rho g h_{\text{centroid}} A_v = \rho g \left(h + \frac{R}{2} \right) (R \cdot w)$$

Scientific Verification: Archimedes' Principle

The fundamental logic that the vertical force F_V exerted by a fluid on a surface is equal to the weight of the fluid volume above it is rooted in the principles laid out by Archimedes of Syracuse (c. 250 BC) in his treatise *On Floating Bodies*.

3.4 Fluid Dynamics: The Bernoulli Equation

3.4.1 Derivation and Strict Assumptions

Applying Newton's Second Law ($\Sigma \vec{F} = m\vec{a}$) along a streamline ($s$-direction):

$$-\frac{\partial P}{\partial s} - \rho g \frac{\partial z}{\partial s} = \rho v \frac{\partial v}{\partial s}$$

Integrating along the streamline with constant ρ gives the Bernoulli Equation (1738):

Bernoulli's Equation

$$P + \frac{1}{2}\rho v^2 + \rho g z = \text{Constant} \tag{39}$$

Strict Assumptions: 1) Steady flow, 2) Incompressible fluid ($\rho =$ cst), 3) Inviscid flow (no friction/shear), 4) Along a single streamline.

3.4.2 Stagnation Point

When fluid hits a blunt object, the streamline dividing the flow stops completely at the Stagnation Point (SP) where $v = 0$.

Using Bernoulli from point 1 to the Stagnation Point (point 2):

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + 0 \implies P_2 = P_1 + \frac{1}{2}\rho v_1^2 = P_{\text{stagnation}}$$

3.4.3 Flow Across Streamlines

In the normal direction (n) towards the center of curvature R_n:

$$-\frac{\partial P}{\partial n} - \rho g \frac{\partial z}{\partial n} = \rho \frac{v^2}{R_n}$$

Expert Analysis: Normal Direction Integration

Integrating this equation across the streamlines (along dn) yields the following exact result:

$$P + \rho g z + \rho \int \frac{v^2}{R_n} dn = \text{Constant}$$

This mathematically demonstrates that pressure increases moving radially outward away from the center of curvature.

4 Week 4 - Applications of Bernoulli & Conservation Laws

4.1 Rectilinear (or Nearly) Streamlines

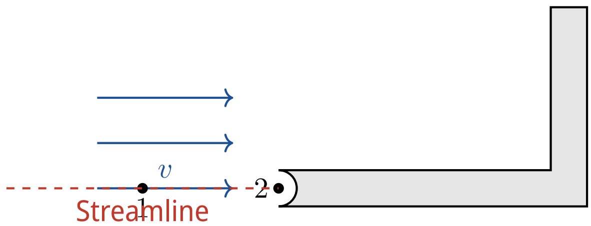
Consider a fluid element moving along a rectilinear streamline. If we neglect dynamic pressure variations transverse to the flow, we recover the hydrostatic pressure distribution:

$$p_1 + \rho g z_1 = p_2 + \rho g z_2$$

$$p_2 - p_1 = -\rho g(z_2 - z_1) = -\rho g h \implies p_1 = p_2 + \rho g h \tag{40}$$

4.2 Example: Pitot-Tube (Speed Sensor)

A Pitot tube is used to measure the local velocity of the fluid by comparing static and stagnation pressures.



Task: Determine the free-stream velocity v. Along the streamline connecting point 1 (free stream) and point 2 (stagnation point), we apply Bernoulli's equation, neglecting gravity ($z_1 \approx z_2$):

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Since point 2 is a stagnation point, $v_2 = 0$. The free stream velocity is $v_1 = v$.

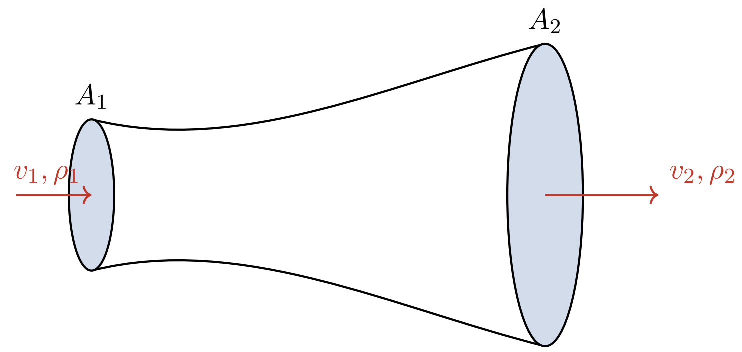
$$p_1 + \frac{1}{2}\rho v^2 = p_2 + 0 \implies p_2 - p_1 = \frac{1}{2}\rho v^2$$

Result: Velocity from Pitot Tube

$$v = \sqrt{\frac{2(p_2 - p_1)}{\rho}}$$

4.3 Mass Conservation

Consider a stream tube with varying cross-sectional area.



- **Mass flow rate** ($\dot{m}$): [kg/s]. $\dot{m} = \rho v A$
- **Volume flow rate** ($\dot{V}$): [m^3/s]. $\dot{V} = vA$

For steady flow through a stream tube, mass is conserved:

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

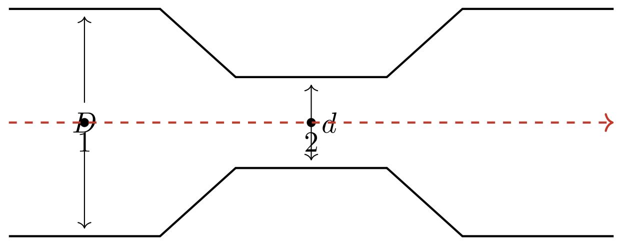
Assuming steady and **incompressible** flow ($\rho_1 = \rho_2$):

Volume Conservation

$$v_1 A_1 = v_2 A_2$$

4.4 Example: Venturi Meter

A Venturi meter measures the flow rate in a pipe by constricting the flow and measuring the pressure drop.



Question: What is the flow rate? Applying Bernoulli's equation along the central streamline (assuming horizontal flow, $z_1 = z_2$):

$$\frac{1}{2}\rho v_1^2 + p_1 = \frac{1}{2}\rho v_2^2 + p_2 \implies v_2^2 - v_1^2 = \frac{2(p_1 - p_2)}{\rho}$$

This equation has two unknowns (v_1 and v_2). We use volume conservation to eliminate v_2:

$$v_1 A_1 = v_2 A_2 \implies v_1\left(\frac{\pi D^2}{4}\right) = v_2\left(\frac{\pi d^2}{4}\right) \implies v_2 = v_1\left(\frac{D}{d}\right)^2$$

Substitute v_2 back into the Bernoulli equation:

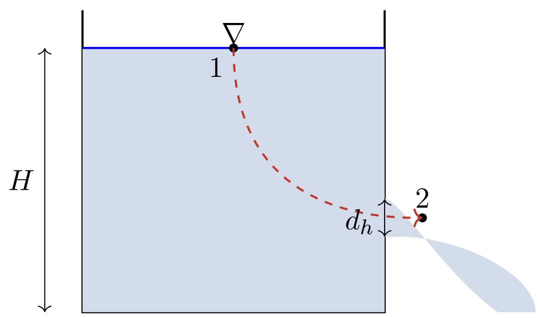
$$v_1^2\left(\frac{D}{d}\right)^4 - v_1^2 = \frac{2(p_1-p_2)}{\rho} \implies v_1^2\left[\left(\frac{D}{d}\right)^4 - 1\right] = \frac{2(p_1-p_2)}{\rho}$$

Result: Venturi Inlet Velocity

$$v_1 = \sqrt{\frac{2(p_1-p_2)}{\rho\left[\left(\frac{D}{d}\right)^4 - 1\right]}}$$

4.5 Example: Free Jet

Consider a large tank filled with fluid up to height H, with a small hole of diameter d at the bottom.



Assume steady or quasi-steady flow with no shear forces. Apply Bernoulli along the streamline from the surface (1) to the exiting jet (2):

$$\frac{1}{2}\rho v_1^2 + \rho g z_1 + p_1 = \frac{1}{2}\rho v_2^2 + \rho g z_2 + p_2$$

Assumptions:

- The tank is large: $v_1 \approx 0$
- Heights: $z_1 = H$, $z_2 = 0$
- Both points are exposed to the atmosphere: $p_1 = p_2 = p_{atm}$

$$\rho g H = \frac{1}{2}\rho v_2^2 \implies v_2 = \sqrt{2gH} \quad \text{(Torricelli's Law)}$$

The ideal volume flow rate is:

$$\dot{V}_{ideal} = v_2 A_2 = \sqrt{2gH}\left(\frac{\pi}{4}d_h^2\right)$$

Vena Contracta Because the streamlines must bend to exit the hole, the actual jet diameter (d_j) contracts to a size smaller than the physical hole diameter (d_h).

Contraction Coefficient (C_c)

$$C_c = \frac{A_j}{A_h} = \left(\frac{d_j}{d_h}\right)^2$$

where $d_j \le d_h$. The true flow rate is scaled by C_c.

5 Week 5 - Chapter D: Diffusion

5.1 D.1 Intro + Diffusion Equation

Definition: Diffusion

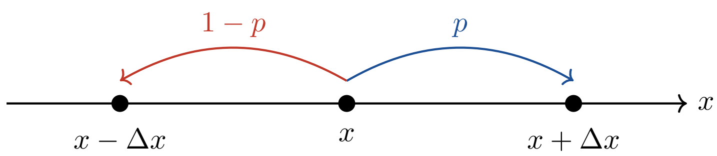
Transport of mass (small particles, molecules, dye) in a fluid *without* macroscopic flow.

Remarks

- Diffusion and convective transport often happen simultaneously.
- Chemical diffusion (mass transport) is mathematically analogous to thermal conduction.

5.1.1 D.1.1 Micro Description of Diffusion

Model: Random Walk on a 1D Lattice Let $n(x,t)$ be the number density of particles at position x at time t. Let Δx be the lattice spacing.



Assume a particle jumps left with probability $1-p$ and right with probability p. If $p = 1/2$, the walk is unbiased; otherwise, it is biased.

To find the change of number density Δn in time Δt, we compute the net flow into position x:

$$\text{Inflow} = n(x+\Delta x, t)(1-p) + n(x-\Delta x, t)p$$
$$\text{Outflow} = n(x,t)[p + (1-p)] = n(x,t)$$

Thus, the change in density is:

$$\Delta n \approx \frac{\partial n}{\partial t}\Delta t = n(x+\Delta x, t)(1-p) + n(x-\Delta x, t)p - n(x,t)$$

Treating $n(x,t)$ as continuous, we perform a Taylor expansion in space up to the second order:

$$n(x+\Delta x, t) \approx n(x) + \Delta x \frac{\partial n}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 n}{\partial x^2}$$
$$n(x-\Delta x, t) \approx n(x) - \Delta x \frac{\partial n}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 n}{\partial x^2}$$

Plugging this into the Δn equation yields:

$$\frac{\partial n}{\partial t}\Delta t = -\Delta x(1-2p)\frac{\partial n}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 n}{\partial x^2}$$

Convection-Diffusion Equation (1D)

$$\frac{\partial n}{\partial t} + v\frac{\partial n}{\partial x} = D\frac{\partial^2 n}{\partial x^2}$$

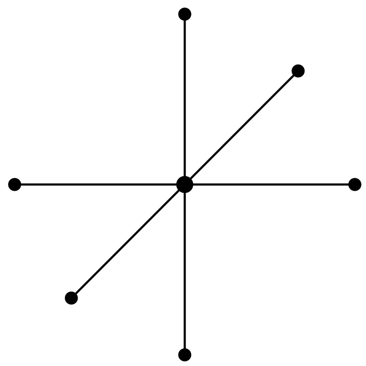
where:

- $D = \frac{\Delta x^2}{2\Delta t}$ is the **diffusion coefficient** $[m^2/s]$.
- $v = (1-2p)\frac{\Delta x}{\Delta t}$ is the **drift velocity**.

For an **unbiased random walk** ($p = 1/2 \implies v = 0$), we recover the standard 1D Diffusion equation:

$$\frac{\partial n}{\partial t} = D\frac{\partial^2 n}{\partial x^2}$$

Extension to 3D Lattice For an unbiased walk on a 3D Cartesian lattice, the jump probability in any of the 6 directions is $p = 1/6$.



Applying the same logic, the equation becomes:

$$\frac{\partial n}{\partial t} = \frac{\Delta x^2}{6\Delta t}\left(\frac{\partial^2 n}{\partial x^2} + \frac{\partial^2 n}{\partial y^2} + \frac{\partial^2 n}{\partial z^2}\right) \implies \frac{\partial n}{\partial t} = D\nabla^2 n \quad \text{with } D = \frac{\Delta x^2}{6\Delta t}$$

Estimating Diffusion Coefficients

- **Self-diffusion** (Kinetic theory): $D \sim \frac{1}{6}\delta\left(\frac{3k_BT}{m}\right)^{1/2}$
- **Suspended particle** (Stokes-Einstein relation, 1905): $\mathcal{D} = \frac{k_BT}{6\pi\mu a}$ for a sphere of radius a in a liquid with dynamic viscosity μ.

—

5.1.2 D.1.2 Continuum Description

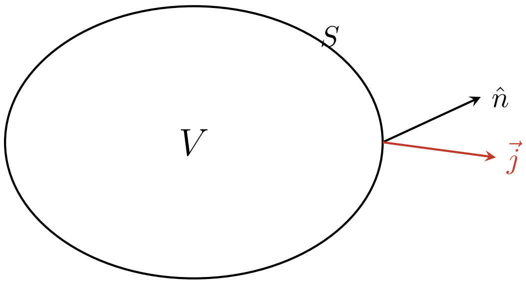
Let $C(\vec{x}, t)$ be the macroscopic concentration field (units: $\frac{1}{m^3}$ or $\frac{\text{mol}}{m^3}$). Observation: Molecules diffuse from high to low concentration.

Definition: Fick's First Law

The particle flux $\vec{j}$ (number of molecules transported through an area per unit time) is proportional to the negative gradient of the concentration:

$$\vec{j} = -D\nabla C$$

Mass Conservation Consider a fixed control volume V bounded by a surface S. The rate of change of mass inside must equal the net flux through the surface:



$$\frac{d}{dt}\int_V C\,dV = -\oint_S \vec{j}\cdot d\vec{S}$$

Using Gauss's Divergence Theorem on the right-hand side:

$$-\oint_S \vec{j}\cdot d\vec{S} = -\int_V \nabla\cdot\vec{j}\,dV$$

For an arbitrary volume V, this implies the differential form of mass conservation:

$$\frac{\partial C}{\partial t} + \nabla\cdot\vec{j} = 0$$

The Diffusion Equation

Substituting Fick's Law ($\vec{j} = -D\nabla C$) into the conservation equation, assuming constant D:

$$\frac{\partial C}{\partial t} = D\nabla^2 C$$

Boundary Conditions

To solve this PDE, appropriate initial and boundary conditions are needed:

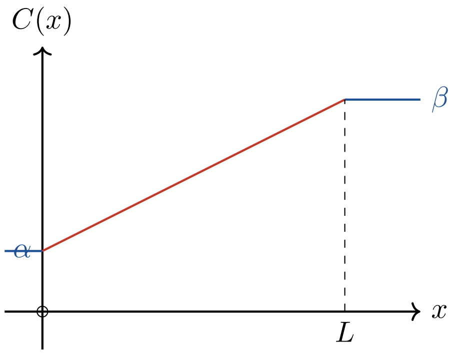
- **Dirichlet (Fixed Concentration):** $C(\text{surface}) = C_0$
- **Neumann (Fixed Flux):** $-D\hat{n}\cdot\nabla C = j_0$
- **Robin (Mixed/Convective):** $j = h(C - C_\infty)$ where h is a mass transfer coefficient.

—

5.2 D.2 Solving the Diffusion Equation (Steady State)

In a steady state, the concentration profile does not change with time, meaning $\frac{\partial C}{\partial t} = 0$. The diffusion equation simplifies to Laplace's equation: $\nabla^2 C = 0$.

5.2.1 Example 1: 1D Thin Pipe

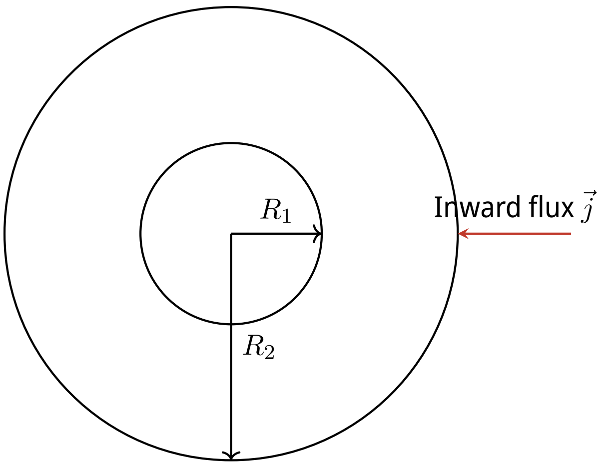


Governing ODE: $D\frac{d^2C}{dx^2} = 0 \implies C(x) = A_1 x + A_2$. Applying Boundary Conditions $C(0) = \alpha$ and $C(L) = \beta$:

$$A_2 = \alpha, \quad A_1 = \frac{\beta - \alpha}{L} \implies C(x) = \frac{\beta - \alpha}{L}x + \alpha$$

Note: The concentration profile is independent of D in 1D steady state, but the steady flux depends on D: $j = -D\frac{\beta-\alpha}{L}$.

5.2.2 Example 2: 3D Spherical Shell



In spherical coordinates with purely radial dependence, the steady-state equation is:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC}{dr}\right) = 0$$

Integrating twice gives the general solution:

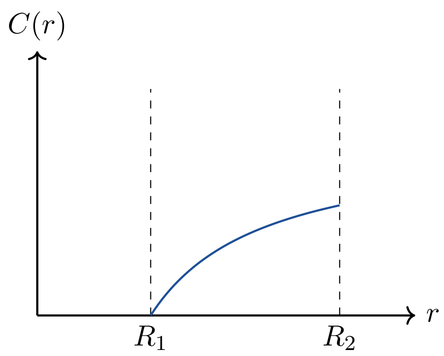
$$r^2\frac{dC}{dr} = A_1 \implies \frac{dC}{dr} = \frac{A_1}{r^2} \implies C(r) = -\frac{A_1}{r} + A_2$$

Applying Boundary Conditions: 1. Inner boundary (Fixed concentration): $C(R_1) = 0 \implies A_2 = \frac{A_1}{R_1}$ 2. Outer boundary (Fixed inward flux α): $D\frac{dC}{dr}\Big|_{R_2} = \alpha \implies D\frac{A_1}{R_2^2} = \alpha \implies A_1 = \frac{\alpha R_2^2}{D}$

Substituting A_1 and A_2 back into the general solution:

Result: Radial Concentration Profile

$$C(r) = \alpha\frac{R_2^2}{D}\left(\frac{1}{R_1} - \frac{1}{r}\right)$$



6 Week 6

6.1 D.3.1 Finite Domain (1D) (Series Expansion)

6.1.1 D.3.1.1 Fixed Concentration Boundaries

The governing Partial Differential Equation (PDE) for 1D time-dependent diffusion is:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} \tag{41}$$

Boundary Conditions (BC) We assume fixed, zero concentration at the physical boundaries $x = 0$ and $x = L$:

$$C(0,t) = C(L,t) = 0 \tag{42}$$

Superposition Principle

The PDE is **linear** and **homogeneous**, and it is subject to **Homogeneous Boundary Conditions (HBC)**. Therefore, if $C_1(x,t)$ and $C_2(x,t)$ are valid solutions, any linear combination is also a valid solution:

$$C_1(x,t) + C_2(x,t) \quad \text{and} \quad \beta C_1(x,t) \quad (\text{for } \beta \in \mathbb{R})$$

Proof via PDE Linearity:

$$\begin{aligned}\frac{\partial}{\partial t}(C_1 + C_2) - D\frac{\partial^2}{\partial x^2}(C_1 + C_2) &= \left(\frac{\partial C_1}{\partial t} - D\frac{\partial^2 C_1}{\partial x^2}\right) + \left(\frac{\partial C_2}{\partial t} - D\frac{\partial^2 C_2}{\partial x^2}\right) \\ &= 0 + 0 = 0\end{aligned}$$

Proof via BCs:

$$(C_1 + C_2)(0,t) = C_1(0,t) + C_2(0,t) = 0 + 0 = 0$$

—

6.1.2 Solution Approach (Method of Separation of Variables)

Because of linearity, the general solution is constructed as an infinite linear combination of base eigenfunctions $\varphi_n(x,t)$:

$$C(x,t) = \sum_{n=1}^{\infty} A_n \varphi_n(x,t) \tag{43}$$

We assume the fundamental solution $\varphi(x,t)$ can be factored into purely spatial and purely temporal functions:

$$\varphi(x,t) = X(x)T(t) \tag{44}$$

Substituting this ansatz into the PDE gives $X(x)T'(t) = DX''(x)T(t)$. Dividing by $DX(x)T(t)$ separates the variables:

$$\frac{T'(t)}{DT(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

Since the LHS strictly depends on t and the RHS strictly depends on x, both must equal a separation constant, denoted $-\lambda$. This yields two coupled Ordinary Differential Equations (ODEs):

$$T'(t) + D\lambda T(t) = 0 \quad \text{(Time dependence)} \tag{45}$$
$$X''(x) + \lambda X(x) = 0 \quad \text{(Spatial dependence)} \tag{46}$$

—

Solving the Spatial Eigenvalue Problem We must solve the 2^{nd} order ODE $X''(x)+\lambda X(x) = 0$ subject to the HBCs $X(0) = X(L) = 0$.

- **Case I ($\lambda < 0$):** $X(x) = \alpha_1 e^{\sqrt{-\lambda}x} + \alpha_2 e^{-\sqrt{-\lambda}x}$. Applying BCs gives $\alpha_1 = \alpha_2 = 0$. Trivial solution.
- **Case II ($\lambda = 0$):** $X(x) = \alpha_1 x + \alpha_2$. Applying BCs gives $\alpha_2 = 0$ and $\alpha_1 L = 0$. Trivial solution.
- **Case III ($\lambda > 0$):** The general solution is a harmonic oscillator form:

  $$X(x) = \alpha_1 \cos(\sqrt{\lambda}x) + \alpha_2 \sin(\sqrt{\lambda}x)$$

  Applying $X(0) = 0 \implies \alpha_1 = 0$.
  Applying $X(L) = 0 \implies \alpha_2 \sin(\sqrt{\lambda}L) = 0$.
  To avoid the trivial solution ($\alpha_2 \neq 0$), the sine term must vanish: $\sin(\sqrt{\lambda}L) = 0$.

This yields the **eigenvalues** and spatial **eigenfunctions**:

Spatial Eigenfunctions

$$\sqrt{\lambda}L = n\pi \quad \implies \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2 \quad \text{for } n \in \mathbb{N}^*$$
$$X_n(x) = \sin\left(\frac{n\pi}{L}x\right)$$

Note: The constant α_n is omitted as it will be absorbed by the Fourier coefficients A_n.

—

Solving the Temporal Problem For a given eigenvalue λ_n, the temporal ODE is a first-order linear decay:

$$T_n'(t) = -D\lambda_n T_n(t) \implies T_n(t) = \beta_n e^{-D\lambda_n t}$$

Substituting λ_n, the temporal amplitude decays exponentially:

$$T_n(t) = e^{-D\left(\frac{n\pi}{L}\right)^2 t} \tag{47}$$

—

6.1.3 Combining the Solutions & Initial Conditions

Multiplying the spatial and temporal components, the general solution satisfying the boundary conditions is:

$$C(x,t) = \sum_{n=1}^{\infty} A_n e^{-D\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L}x\right) \tag{48}$$

Determining the Fourier Coefficients A_n At $t = 0$, the solution must match the given initial concentration profile $C_0(x)$:

$$C(x,0) = C_0(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right)$$

To isolate A_n, we multiply both sides by $\sin\left(\frac{m\pi}{L}x\right)$ (for integer m) and integrate over the domain $[0, L]$:

$$\int_0^L C_0(x) \sin\left(\frac{m\pi}{L}x\right) dx = \sum_{n=1}^{\infty} A_n \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx$$

Expert Analysis: Fourier Orthogonality

Using the trigonometric identity $\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$, the integral of the product of two sine waves over the domain evaluates to:

$$\int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx = \begin{cases} 0 & \text{if } n \neq m \\ \frac{L}{2} & \text{if } n = m \end{cases}$$

Due to this orthogonality, the infinite sum collapses to a single non-zero term when $n = m$:

$$\int_0^L C_0(x) \sin\left(\frac{n\pi}{L}x\right) dx = A_n\left(\frac{L}{2}\right) \implies A_n = \frac{2}{L}\int_0^L C_0(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Result: Complete Analytical Solution

For an arbitrary initial condition $C_0(x)$, the full space-time solution is the Fourier sine series:

$$C(x,t) = \sum_{n=1}^{\infty} \underbrace{\left[\frac{2}{L}\int_0^L C_0(\xi) \sin\left(\frac{n\pi}{L}\xi\right) d\xi\right]}_{\text{Fourier Coefficient } A_n} \cdot \underbrace{e^{-D\left(\frac{n\pi}{L}\right)^2 t}}_{\text{Temporal Decay}} \cdot \underbrace{\sin\left(\frac{n\pi}{L}x\right)}_{\text{Spatial Mode}} \tag{49}$$

Note: The dummy integration variable has been set to ξ to strictly separate it from the spatial coordinate x.

7 Week 7 - Diffusion and Partial Differential Equations

7.1 Non-Homogeneous Boundary Conditions (BC)

Concept: The idea is to decompose the full solution into a steady-state part that satisfies the non-homogeneous boundary conditions and a transient part that satisfies homogeneous boundary conditions. As $t \to \infty$, the transient contribution decays and the solution converges to the steady-state profile.

The governing one-dimensional diffusion (or heat) equation is:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} \tag{50}$$

Boundary Conditions (BC):

- $C(0,t) = C_1$
- $C(L,t) = C_2$

Initial Condition (IC):

- $C(x,0) = C_0(x)$

7.1.1 Step A: Solve for the Steady State

Define the steady-state solution by

$$C_f(x) = \lim_{t\to\infty} C(x,t) \tag{51}$$

At steady state, the time derivative vanishes:

$$\frac{\partial C_f}{\partial t} = 0 \qquad \Longrightarrow \qquad 0 = D\frac{\partial^2 C_f}{\partial x^2} \tag{52}$$

Hence,

$$C_f(x) = C_1 + \frac{C_2 - C_1}{L}x \tag{53}$$

Critical Analysis: Limits of the Steady-State Approximation

Mathematical rigor vs. physical reality: The formal definition $C_f(x) = \lim_{t\to\infty} C(x,t)$ refers to an infinite-time limit. In practice, a system is considered close to steady state when the Fourier number $Fo = \frac{Dt}{L^2}$ becomes sufficiently large, typically $Fo > 0.2$ as a rough engineering estimate. In addition, the linear profile above assumes that the diffusivity D is constant. If D depends on concentration, temperature, or composition, then the PDE becomes non-linear and the steady-state profile is no longer necessarily linear.

7.1.2 Step B: Define the Transient Component

Introduce the deviation from steady state:

$$\tilde{C}(x,t) := C(x,t) - C_f(x) \tag{54}$$

Then $\tilde{C}$ satisfies the homogeneous PDE

$$\frac{\partial \tilde{C}}{\partial t} = D\frac{\partial^2 \tilde{C}}{\partial x^2} \tag{55}$$

The transformed homogeneous boundary conditions are:

- $\tilde{C}(0,t) = C(0,t) - C_f(0) = C_1 - C_1 = 0$
- $\tilde{C}(L,t) = C(L,t) - C_f(L) = C_2 - C_2 = 0$

The transformed initial condition is:

- $\tilde{C}(x,0) = C_0(x) - C_f(x)$

7.1.3 Step C: Superposition

The full solution is reconstructed as

$$C(x,t) = C_f(x) + \tilde{C}(x,t) \tag{56}$$

7.2 Time-Dependent Diffusion (1D Series Expansion)

The governing PDE for one-dimensional time-dependent diffusion is:

Definition: The Diffusion Equation

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} \tag{57}$$

Boundary Conditions (BC): Fixed concentration
Assume zero concentration at both boundaries:

$$C(0,t) = C(L,t) = 0 \tag{58}$$

Superposition Principle

The PDE is linear and homogeneous, and the boundary conditions are also homogeneous. Therefore, if $C_1(x,t)$ and $C_2(x,t)$ are solutions, then

$$C_1(x,t) + C_2(x,t) \qquad \text{and} \qquad \beta C_1(x,t), \quad \beta \in \mathbb{R}$$

are also solutions.

Why is this true? (Superposition Principle)

$$\text{PDE:}\quad \frac{\partial}{\partial t}(C_1 + C_2) - D\frac{\partial^2}{\partial x^2}(C_1 + C_2) = \left(\frac{\partial C_1}{\partial t} - D\frac{\partial^2 C_1}{\partial x^2}\right) + \left(\frac{\partial C_2}{\partial t} - D\frac{\partial^2 C_2}{\partial x^2}\right)$$
$$= 0 + 0 = 0$$

and the boundary conditions are preserved because

$$(C_1 + C_2)(0,t) = C_1(0,t) + C_2(0,t) = 0 + 0 = 0$$
$$(C_1 + C_2)(L,t) = C_1(L,t) + C_2(L,t) = 0 + 0 = 0$$

7.2.1 Solution Approach (Linearity)

Step 1: General solution representation
Because of linearity, the solution can be written as a superposition of basis functions:

$$C(x,t) = \sum_{n=1}^{\infty} A_n \varphi_n(x,t) \tag{59}$$

where the coefficients A_n are determined by the initial condition.

Step 2: Separation of variables
Assume

$$C(x,t) = X(x)T(t) \tag{60}$$

Substituting into the PDE gives

$$X(x)\frac{dT(t)}{dt} = D\,T(t)\frac{d^2X(x)}{dx^2}$$

Dividing by $D\,X(x)T(t)$ yields

$$\underbrace{\frac{1}{D}\frac{T'(t)}{T(t)}}_{\text{depends only on } t} = \underbrace{\frac{X''(x)}{X(x)}}_{\text{depends only on } x} = -\lambda \tag{61}$$

Since the left-hand side depends only on t and the right-hand side only on x, both must be equal to a constant $-\lambda$.

This leads to two coupled ODEs:

$$T'(t) + D\lambda T(t) = 0 \tag{62}$$
$$X''(x) + \lambda X(x) = 0 \tag{63}$$

7.2.2 Solving the Spatial Problem

We now solve

$$X''(x) + \lambda X(x) = 0 \tag{64}$$

subject to

$$X(0) = X(L) = 0 \tag{65}$$

Definition: Evaluating the different cases for λ

Case I: $\lambda < 0$
The solution has the form

$$X(x) = \alpha_1 e^{\sqrt{-\lambda}\,x} + \alpha_2 e^{-\sqrt{-\lambda}\,x}$$

Applying the boundary conditions forces $\alpha_1 = \alpha_2 = 0$, giving only the trivial solution.

Case II: $\lambda = 0$
The solution becomes

$$X(x) = \alpha_1 x + \alpha_2$$

Applying the boundary conditions again yields only the trivial solution.

Case III: $\lambda > 0$
The solution is

$$X(x) = \alpha_1 \cos(\sqrt{\lambda}\,x) + \alpha_2 \sin(\sqrt{\lambda}\,x)$$

Applying the boundary conditions gives

$$X(0) = \alpha_1 = 0$$
$$X(L) = \alpha_2 \sin(\sqrt{\lambda}\,L) = 0$$

To avoid the trivial solution, we require

$$\sin(\sqrt{\lambda}\,L) = 0$$

which implies

$$\sqrt{\lambda}\,L = k\pi, \qquad k \in \mathbb{Z}^+$$

Therefore, the eigenvalues are

$$\lambda_k = \left(\frac{k\pi}{L}\right)^2, \qquad k \in \{1, 2, 3, \dots\} \tag{66}$$

and the corresponding eigenfunctions are

$$X_k(x) = \sin\left(\frac{k\pi}{L}x\right) \tag{67}$$

Note: The amplitude factor can be absorbed into the global coefficient A_k.

7.2.3 Solving the Temporal Problem

For a fixed eigenvalue

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 \tag{68}$$

the time equation becomes

$$T_n'(t) = -D\lambda_n T_n(t)$$

whose solution is

$$T_n(t) = \beta_n e^{-D\left(\frac{n\pi}{L}\right)^2 t} \tag{69}$$

7.2.4 Combining the Spatial and Temporal Solutions

The basis solutions are therefore

$$\varphi_n(x,t) = e^{-D\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L}x\right) \tag{70}$$

and the general solution satisfying the homogeneous boundary conditions is

$$C(x,t) = \sum_{n=1}^{\infty} A_n e^{-D\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L}x\right) \tag{71}$$

7.2.5 Step 3: Determine the coefficients $\{A_n\}$ from the initial condition

At $t = 0$,

$$C(x,0) = C_0(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) \tag{72}$$

To determine the coefficients, multiply both sides by $\sin\left(\frac{m\pi}{L}x\right)$ and integrate from 0 to L:

$$\int_0^L C_0(x) \sin\left(\frac{m\pi}{L}x\right)\, dx = \sum_{n=1}^{\infty} A_n \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right)\, dx$$

Orthogonality Property

$$\int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right)\, dx = \begin{cases} 0 & \text{if } n \neq m \\ \frac{L}{2} & \text{if } n = m \end{cases}$$

Trigonometric identity used:

$$\sin(a)\sin(b) = \frac{1}{2}\left[\cos(a-b) - \cos(a+b)\right]$$

By orthogonality, only the term $n = m$ remains:

$$\int_0^L C_0(x) \sin\left(\frac{n\pi}{L}x\right)\, dx = A_n \frac{L}{2}$$

Hence,

$$A_n = \frac{2}{L} \int_0^L C_0(x) \sin\left(\frac{n\pi}{L}x\right)\, dx \tag{73}$$

Result: Final Full Solution

For any initial condition $C_0(x)$ that admits a Fourier sine expansion, the complete solution is

$$C(x,t) = \sum_{n=1}^{\infty} \left[\frac{2}{L} \int_0^L C_0(\tilde{x}) \sin\left(\frac{n\pi}{L}\tilde{x}\right)\, d\tilde{x}\right] e^{-D\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L}x\right) \tag{74}$$

7.3 Visualizing Flow Patterns

Fluid kinematics describes motion without yet invoking conservation laws. The three main geometric ways to represent a flow are streamlines, pathlines, and streaklines.

1. **Streamlines:** Curves tangent to the instantaneous velocity field at a given time.

$$\frac{dy}{dx} = \frac{v}{u}, \qquad \frac{dz}{dx} = \frac{w}{u}, \qquad \frac{dz}{dy} = \frac{w}{v} \qquad \Longrightarrow \qquad \frac{d\vec{r}_{SL}}{ds} \times \vec{V} = \vec{0} \tag{75}$$

2. **Pathlines:** The actual trajectory followed by one specific fluid particle through time.

$$\frac{d\vec{r}_p}{dt} = \vec{V}(\vec{r}_p(t), t) \qquad \Longrightarrow \qquad \frac{dx}{dt} = u,\ \frac{dy}{dt} = v,\ \frac{dz}{dt} = w \tag{76}$$

3. **Streaklines:** The locus of all particles that have passed through a given point $\vec{r}_0$ at previous times $\tau < t$.

Key Result: In steady flow, where $\partial\vec{V}/\partial t = 0$, streamlines, pathlines, and streaklines coincide.

7.3.1 Example: The Sprinkler (Unsteady Case)

Consider the unsteady velocity field

$$\vec{V} = U_0 \sin\left[\omega\left(t - \frac{y}{v_0}\right)\right] \hat{\imath} + v_0 \hat{\jmath} \tag{77}$$

- **Streamlines at $t = 0$:**

$$x(y) = \frac{U_0}{\omega}\left(\cos\left(\frac{\omega y}{v_0}\right) - 1\right) \tag{78}$$

  so the streamline shape is sinusoidal.

- **Pathline for a particle starting at the origin at $t = 0$:**

$$x(t) = 0, \qquad y(t) = v_0 t \tag{79}$$

  which is a straight vertical line.

This example shows that in unsteady flow, streamlines and pathlines are generally different objects.

7.4 Acceleration Field

The acceleration of a fluid particle is the total derivative of the velocity field:

$$\vec{a} = \frac{D\vec{V}}{Dt} = \underbrace{\frac{\partial\vec{V}}{\partial t}}_{\text{Local acceleration}} + \underbrace{(\vec{V}\cdot\vec{\nabla})\vec{V}}_{\text{Convective acceleration}} \tag{80}$$

Technical Analysis: The Material Derivative

The operator $\frac{D}{Dt}$ is fundamental in fluid mechanics because it gives the rate of change experienced by a moving fluid particle. The convective term $(\vec{V}\cdot\vec{\nabla})\vec{V}$ is non-linear and is a major source of complex flow behavior, including mechanisms that contribute to turbulence. It measures the change in velocity caused by a particle moving into a region where the flow field is different, even when the flow is steady in time.

7.4.1 Example: The Diffuser

For a steady one-dimensional flow

$$u(x) = v_0\left(1-\frac{x}{L}\right) \tag{81}$$

the local acceleration term vanishes because the flow is steady, so only the convective contribution remains:

$$a_x = u\frac{\partial u}{\partial x} = v_0\left(1-\frac{x}{L}\right)\left(-\frac{v_0}{L}\right) = -\frac{v_0^2}{L}\left(1-\frac{x}{L}\right) \tag{82}$$

The negative sign indicates deceleration, which is consistent with the widening geometry of a diffuser.

7.5 Velocity Fields: Eulerian vs. Lagrangian Descriptions

Once the motion has been visualized geometrically, it is useful to introduce the two standard ways of describing it mathematically.

Eulerian Description: Focuses on fixed points in space and describes the velocity as a field:

$$\vec{V}(\vec{r},t) = u\hat{i} + v\hat{j} + w\hat{k} \tag{83}$$

Lagrangian Description: Follows individual fluid particles and describes their motion from an initial position:

$$\vec{V}(\vec{r}_0,t) \qquad \text{where } \vec{r}_0 \text{ is the particle's initial position} \tag{84}$$

Structural Foundation: Eulerian vs. Lagrangian Comparison

Eulerian formulations (Leonhard Euler, 1757) are the standard framework for most continuum fluid mechanics and Computational Fluid Dynamics (CFD), because they describe the flow as fields defined everywhere in space. Lagrangian formulations (Joseph-Louis Lagrange, 1788) are indispensable when one must follow individual particles, droplets, interfaces, or solid bodies.

Feature	**Eulerian (Field)**	**Lagrangian (Particle)**
Independent Variables	Position $\vec{r}$, Time t	Initial Position $\vec{r}_0$, Time t
Mass Tracking	Flux through a fixed volume	Constant identity of mass
Differentiation	Requires the Material Derivative	Standard time derivative

8 Week 8 - System, Control Volume, and Reynolds Transport Theorem

This section closes the chapter sequence by linking the Lagrangian and Eulerian viewpoints and showing how conservation laws are written for control volumes.

8.1 System vs. Control Volume

- **System:** A fixed amount of mass. Its boundary may move and deform with the material. This is the natural **Lagrangian** viewpoint.
- **Control Volume (CV):** A region in space through which mass may enter or leave. This is the natural **Eulerian** viewpoint.

The purpose of the Reynolds Transport Theorem is to connect these two descriptions.

8.2 Reynolds Transport Theorem (RTT)

Formal derivation: Munson 4.4.1

Consider a system at time $t = t_0$ such that the control volume and the system coincide:

$$CV \text{ at } t = t_0 = SYS \tag{85}$$

Let

- B be an extensive quantity
- b be the corresponding intensive quantity defined by

$$b = \frac{B}{m} \tag{86}$$

The amount of B inside the control volume and inside the system can be written as

$$B_{CV} = \int_{CV} \rho b \, dV, \qquad B_{SYS} = \int_{SYS} \rho b \, dV \tag{87}$$

The rate of change of B for the system is related to the control-volume description by

$$\frac{DB_{SYS}}{Dt} = \frac{\partial B_{CV}}{\partial t} + \dot{B}_{out} - \dot{B}_{in} \tag{88}$$

The flux terms are

- Flux of B into the control volume:

$$\dot{B}_{in} = b_{in}\dot{m}_{in} = b_{in}\rho_{in}A_{in}V_{in} \tag{89}$$

- Flux of B out of the control volume:

$$\dot{B}_{out} = b_{out}\dot{m}_{out} = b_{out}\rho_{out}A_{out}V_{out} \tag{90}$$

Simple Version of RTT

$$\left(\frac{DB}{Dt}\right)_{SYS} = \left(\frac{\partial B}{\partial t}\right)_{CV} + (b\rho VA)_{out} - (b\rho VA)_{in}$$

8.2.1 General Form of RTT

For a general surface, the mass flow rate through an area element is

$$\dot{m} = \int_A \rho\vec{V} \cdot \hat{n}\, dA \tag{91}$$

Therefore, the net rate at which the property B crosses the control surface is

$$\dot{B} = \int_{CS} b\rho V_n\, dA = \dot{B}_{out} - \dot{B}_{in} \tag{92}$$

where

- $\hat{n}$ is the outward unit normal vector
- dA is an area element on the control surface
- dV is a volume element in the control volume
- $V_n = \vec{V} \cdot \hat{n}$ is the normal component of velocity

Since $CV = SYS$ at the instant $t = t_0$, the general form of the Reynolds Transport Theorem is

Reynolds Transport Theorem (General Form)

$$\underbrace{\left(\frac{DB}{Dt}\right)_{SYS}}_{\text{Lagrangian system}} = \underbrace{\frac{\partial}{\partial t}\int_{CV} b\rho\, dV}_{\text{Eulerian volume contribution}} + \underbrace{\int_{CS} b\rho(\vec{V} \cdot \hat{n})\, dA}_{\text{Eulerian surface contribution}}$$

Notes:

- For **steady flow**,

$$\frac{\partial}{\partial t}\int_{CV} b\rho\, dV = 0 \tag{93}$$

 so that

$$\frac{DB_{SYS}}{Dt} = \int_{CS} b\rho(\vec{V} \cdot \hat{n})\, dA \tag{94}$$

- For a **moving, non-deforming control volume**, one uses the relative velocity

$$\vec{W} = \vec{V} - \vec{V}_{CV} \tag{95}$$

- In practical calculations, the closed-surface integral is often split into separate inlet and outlet contributions.

8.3 Control Volume Approach to Conservation Laws

Using RTT, the main Lagrangian conservation laws can be rewritten in Eulerian control-volume form:

1. Mass conservation
2. Newton's second law
3. Energy conservation (First Law of Thermodynamics)

8.4 Mass Conservation

For mass conservation, the extensive quantity is the mass itself:

$$B = M \tag{96}$$

Hence the associated intensive quantity is

$$b = 1 \tag{97}$$

Substituting into the general RTT gives

$$\left(\frac{DM}{Dt}\right)_{SYS} = \frac{\partial}{\partial t}\int_{CV} \rho \, dV + \int_{CS} \rho(\vec{V} \cdot \hat{n}) \, dA \tag{98}$$

In words:

> Rate of change of the mass of the coincident system = rate of change of mass inside the control volume + net mass flow rate through the control surface.

For a closed system, mass is conserved, so

$$\left(\frac{DM}{Dt}\right)_{SYS} = 0 \tag{99}$$

This gives the integral continuity equation:

Continuity Equation

$$0 = \frac{\partial}{\partial t}\int_{CV} \rho \, dV + \int_{CS} \rho(\vec{V} \cdot \hat{n}) \, dA$$

8.4.1 Example: Nozzle (Stationary Flow)

Consider a nozzle with:

- **Inlet (Area A_0):** uniform velocity profile V_0, radius R_0
- **Outlet (Area A_1):** parabolic velocity profile

$$V(r) = V_m \left(1 - \frac{r^2}{R_1^2}\right) \tag{100}$$

  with outlet radius R_1

Assumptions:

- Steady flow
- Constant density, $\rho = \text{const}$

Apply the continuity equation:

$$\frac{\partial}{\partial t} \int_{CV} \rho \, dV + \int_{CS} \rho(\vec{V} \cdot \hat{n}) \, dA = 0 \tag{101}$$

Because the flow is steady, the accumulation term vanishes:

$$\frac{\partial}{\partial t} \int_{CV} \rho \, dV = 0 \tag{102}$$

Therefore,

$$\int_{CS} \rho(\vec{V} \cdot \hat{n}) \, dA = 0 \tag{103}$$

and since ρ is constant,

$$\int_{CS} (\vec{V} \cdot \hat{n}) \, dA = 0 \tag{104}$$

Split the control-surface integral into inlet and outlet contributions:

$$\int_{A_0} (\vec{V} \cdot \hat{n}) \, dA + \int_{A_1} (\vec{V} \cdot \hat{n}) \, dA = 0 \tag{105}$$

Inlet contribution:
At the inlet, the outward normal points opposite to the incoming flow, so

$$\vec{V} \cdot \hat{n} = -V_0 \tag{106}$$

Hence,

$$\int_{A_0} (\vec{V} \cdot \hat{n}) \, dA = -V_0 \int_{A_0} dA = -V_0 \pi R_0^2 \tag{107}$$

Outlet contribution:
Using polar coordinates, the area element is

$$dA = 2\pi r \, dr \tag{108}$$

Thus,

$$\begin{aligned}\int_{A_1} (\vec{V} \cdot \hat{n})\, dA &= \int_0^{R_1} V_m \left(1 - \frac{r^2}{R_1^2}\right) (2\pi r)\, dr \\ &= 2\pi V_m \left[\frac{r^2}{2} - \frac{r^4}{4R_1^2}\right]_0^{R_1} \\ &= 2\pi V_m \left(\frac{R_1^2}{2} - \frac{R_1^2}{4}\right) \\ &= \frac{\pi}{2} V_m R_1^2\end{aligned}$$

Combine both contributions:

$$0 = -V_0 \pi R_0^2 + \frac{\pi}{2} V_m R_1^2 \tag{109}$$

Solving for the maximum outlet velocity gives

Key Result

$$V_m = 2V_0 \left(\frac{R_0}{R_1}\right)^2$$

9 Week 9 - RTT and Newton's 2nd Law

We begin by applying the Reynolds Transport Theorem (RTT) to the conservation of linear momentum. Let the extensive property be the linear momentum of the system, $B = m\vec{v}$, and the intensive property be the velocity, $\beta = \vec{v}$.

Newton's second law for a system (a fixed mass) states that the rate of change of linear momentum equals the sum of external forces:

$$\left(\frac{DB}{Dt}\right)_{\text{sys}} = \sum \vec{F}_{\text{sys}} \tag{110}$$

Applying the RTT for a control volume (CV) with a control surface (CS):

$$\underbrace{\frac{\partial}{\partial t}\int_{\text{CV}} \rho\vec{v}\, dV}_{\substack{\text{Rate of change of}\\ \text{linear momentum}\\ \text{in CV}}} + \underbrace{\int_{\text{CS}} \rho\vec{v}(\vec{v}\cdot\hat{n})\, dA}_{\substack{\text{Flux of linear}\\ \text{momentum through CS}}} = \sum \vec{F}_{\text{CV}} \tag{111}$$

This is the fundamental integral form of the momentum equation.

9.1 Simplifications for Practical Problems

Assuming a fixed, non-deforming Control Volume in an inertial reference frame, we can simplify the flux integral if the control surface consists of N discrete flat areas (inlets/outlets):

$$\int_{\text{CS}} (\dots)dA = \int_{A_1} (\dots)dA + \int_{A_2} (\dots)dA + \dots + \int_{A_N} (\dots)dA, \quad N < \infty \tag{112}$$

If the velocity $\vec{v}$ is **uniform** over an area A and parallel to the normal $\hat{n}$:

$$\int_A \rho\vec{v}(\vec{v}\cdot\hat{n})\, dA = \rho\vec{v}(\vec{v}\cdot\hat{n})A = \rho\vec{v}V_n A = \dot{m}\vec{v} \tag{113}$$

where $\dot{m}$ is the mass flow rate.

Steady State & Continuity

If the flow is **steady**, the local derivative term vanishes:

$$\frac{\partial}{\partial t}\int_{\text{CV}} \rho\vec{v}\, dV = 0 \tag{114}$$

For a stream tube with one inlet (1) and one outlet (2), assuming steady uniform flow, the continuity equation requires:

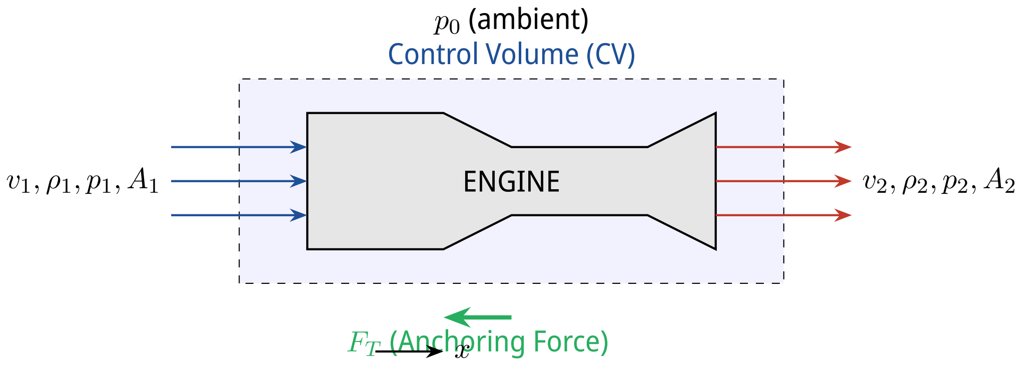
$$\dot{m}_1 = \dot{m}_2 = \dot{m} \quad \implies \quad \int_{\text{CS}} \rho(\vec{v}\cdot\hat{n})\, dA = 0 \tag{115}$$

The momentum equation simplifies to:

$$\sum \vec{F} = \dot{m}_2\vec{v}_2 - \dot{m}_1\vec{v}_1 = \dot{m}(\vec{v}_2 - \vec{v}_1) \tag{116}$$

9.2 Example: Engine Thrust & Anchoring Force

Consider a jet engine. We wish to find the anchoring force F_T required to hold the engine stationary.



Assumptions: Steady flow, uniform velocity profiles. Note that velocities in the x-direction are defined as u_1 and u_2.

Evaluating the momentum equation in the x-direction:

$$-\rho_1 u_1(u_1)A_1 + \rho_2 u_2(u_2)A_2 = \sum F_x \quad \Longrightarrow \quad \dot{m}(u_2 - u_1) = \sum F_x \tag{117}$$

The net force $\sum F_x$ includes the pressure acting on the control surface boundaries and the anchoring force F_T. Pressure acts inwards on the CV:

$$\sum F_x = p_1 A_1 + F_T - p_2 A_2 - p_0(A_1 - A_2) \tag{118}$$

Solving for the anchoring force F_T:

$$F_T = \dot{m}(u_2 - u_1) - (p_1 - p_0)A_1 + (p_2 - p_0)A_2 \tag{119}$$

Expert Analysis: Momentum Flux in Non-Uniform and Deforming Systems

1. Deforming Control Volumes: If the boundaries of the control volume are moving with velocity $\vec{v}_{cs}$, the RTT must be modified to use the relative velocity $\vec{w} = \vec{v} - \vec{v}_{cs}$ for the flux term:

$$\left(\frac{DB}{Dt}\right)_{\text{sys}} = \frac{\partial}{\partial t}\int_{\text{CV}} \rho\beta \, dV + \int_{\text{CS}} \rho\beta(\vec{w}\cdot\hat{n}) \, dA \tag{120}$$

Crucially, $\frac{\partial}{\partial t}$ cannot be moved inside the integral unless the limits of integration (the CV shape) are time-invariant. Even if the internal flow field is "steady" in a relative sense, a deforming boundary implies local unsteadiness.

2. Boussinesq Momentum Correction Factor: The assumption that $\int_A \rho u^2 dA = \dot{m}\bar{u}$ is strictly true *only* for inviscid, plug flow. Real flows have boundary layers. To maintain rigor while using 1D average velocities, a momentum correction factor β_m is introduced:

$$\int_A \rho u^2 dA = \beta_m \dot{m}\bar{u} \quad \text{where} \quad \beta_m = \frac{1}{A}\int_A \left(\frac{u}{\bar{u}}\right)^2 dA \tag{121}$$

For fully developed laminar pipe flow, $\beta_m = 1.33$. For high-Reynolds turbulent flow, $\beta_m \approx 1.01$ to 1.05. Therefore, the uniform flow assumption ($\beta_m = 1$) is an acceptable approximation for turbulent flows but introduces significant error in laminar regimes.

9.3 RTT for the 1st Law of Thermodynamics

9.3.1 General Situation

Let the extensive property be total energy, $B = E$. The corresponding intensive property (specific energy) is $\beta = e = E/m$.

$$e = \tilde{u} + \frac{1}{2}v^2 + gz \tag{122}$$

where $\tilde{u}$ is specific internal energy, $\frac{1}{2}v^2$ is specific kinetic energy, and gz is specific potential energy (assuming gravity acts in the $-z$ direction, $\vec{g} = -g\hat{k}$).

The 1st Law of Thermodynamics states:

$$\frac{DE_{\text{sys}}}{Dt} = \dot{Q}_{\text{net, in}} + \dot{W}_{\text{net, in}} \tag{123}$$

Where $\dot{Q}_{\text{net, in}}$ is the net heat transfer rate *into* the system, and $\dot{W}_{\text{net, in}}$ is the net power (work rate) done *on* the system.

Applying RTT:

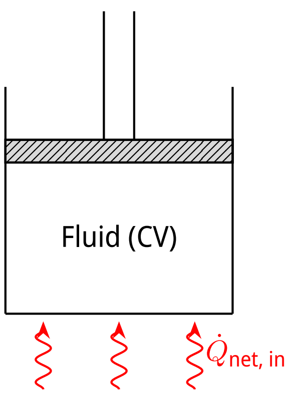
$$\frac{\partial}{\partial t}\int_{\text{CV}} e\rho\, dV + \int_{\text{CS}} e\rho(\vec{v}\cdot\hat{n})\, dA = \dot{Q}_{\text{net, in}} + \dot{W}_{\text{net, in}} \tag{124}$$

9.3.2 Types of Power ($\dot{W}$)

Work requires a force acting over a displacement. Power is force times velocity. We typically divide power into two categories for fluids:

1. **Shaft Power ($\dot{W}_{\text{shaft}}$):** Power transmitted across the control surface by rotating shafts (e.g., pumps adding energy, turbines extracting energy). $\dot{W}_{\text{shaft}} = \tau_{\text{shaft}} \cdot \omega$.
2. **Flow Work / Normal Stress Work ($\dot{W}_{\text{normal}}$):** The work done by pressure forces at the control surfaces to push the fluid into/out of the control volume.

$$\dot{W}_{\text{normal}} = -\int_{\text{CS}} p(\vec{v}\cdot\hat{n})\, dA \tag{125}$$



Substituting $\dot{W}_{\text{net}} = \dot{W}_{\text{shaft}} - \int_{\text{CS}} p(\vec{v} \cdot \hat{n}) dA$ into the RTT formulation and grouping the area integrals:

$$\frac{\partial}{\partial t} \int_{\text{CV}} e\rho \, dV + \int_{\text{CS}} \underbrace{\left(\tilde{u} + \frac{1}{2}v^2 + gz + \frac{p}{\rho} \right)}_{h = \text{enthalpy+mechanical energy}} \rho(\vec{v} \cdot \hat{n}) \, dA = \dot{Q}_{\text{net, in}} + \dot{W}_{\text{shaft, in}} \tag{126}$$

This is the general Energy Equation for a control volume.

Expert Analysis: Derivation of Flow Work from the Cauchy Stress Tensor

The standard derivation groups pressure p into the energy equation, but mathematically, this originates from the surface traction vector $\vec{t} = \boldsymbol{\sigma} \cdot \hat{n}$. The total Cauchy stress tensor is $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}$ (where $\boldsymbol{\tau}$ is the viscous shear stress tensor).
The mechanical power done by the fluid surroundings on the control surface is:

$$\dot{W}_{\text{surface}} = \int_{\text{CS}} (\vec{t} \cdot \vec{v}) \, dA = \int_{\text{CS}} ((-p\mathbf{I} + \boldsymbol{\tau}) \cdot \hat{n}) \cdot \vec{v} \, dA = \int_{\text{CS}} -p(\vec{v} \cdot \hat{n}) \, dA + \int_{\text{CS}} (\boldsymbol{\tau} \cdot \hat{n}) \cdot \vec{v} \, dA \tag{127}$$

In most pipe flows, the control surfaces are perpendicular to the velocity ($\vec{v} \parallel \hat{n}$). Because fluid adheres to stationary pipe walls (no-slip condition, $\vec{v} = 0$), viscous work at solid boundaries is identically zero. Viscous work at the inlets/outlets is typically negligible because velocity gradients in the flow direction ($\partial u / \partial x$) are incredibly small compared to radial gradients. Thus, $\int_{\text{CS}} (\boldsymbol{\tau} \cdot \hat{n}) \cdot \vec{v} \, dA \approx 0$, leaving only the $-p(\vec{v} \cdot \hat{n})$ term.

9.3.3 Simplified Steady Flow Energy Equation

Assuming steady flow ($\partial / \partial t = 0$), a single inlet (in) and single outlet (out), and uniform properties over the cross-subsections:

$$\left(\tilde{u} + \frac{1}{2}v^2 + gz + \frac{p}{\rho} \right)_{\text{out}} \dot{m} - \left(\tilde{u} + \frac{1}{2}v^2 + gz + \frac{p}{\rho} \right)_{\text{in}} \dot{m} = \dot{Q}_{\text{net}} + \dot{W}_{\text{shaft}} \tag{128}$$

9.3.4 Extension to Bernoulli Equation (Mechanical Energy)

Divide the equation by the mass flow rate $\dot{m}$ to express everything per unit mass:

$$\left(\frac{1}{2}v^2 + gz + \frac{p}{\rho} \right)_{\text{out}} - \left(\frac{1}{2}v^2 + gz + \frac{p}{\rho} \right)_{\text{in}} = q_{\text{net}} + w_{\text{shaft}} - (\tilde{u}_{\text{out}} - \tilde{u}_{\text{in}}) \tag{129}$$

where $q_{\text{net}} = \dot{Q}_{\text{net}} / \dot{m}$ and $w_{\text{shaft}} = \dot{W}_{\text{shaft}} / \dot{m}$.

We define the thermodynamic **loss** term (due to internal friction and viscosity) as:

$$\text{loss} = (\tilde{u}_{\text{out}} - \tilde{u}_{\text{in}}) - q_{\text{net}} \tag{130}$$

Thus, we obtain the **Extended Bernoulli Equation**:

Extended Bernoulli with Head Losses

$$\left(\frac{1}{2}v^2 + gz + \frac{p}{\rho}\right)_{\text{out}} - \left(\frac{1}{2}v^2 + gz + \frac{p}{\rho}\right)_{\text{in}} = w_{\text{shaft}} - \text{loss} \tag{131}$$

This differs from the classical Bernoulli equation ($w_{\text{shaft}} - \text{loss} = 0$) by strictly accounting for mechanical work added (e.g., pumps) and mechanical energy dissipated into heat via viscosity (losses).

9.3.5 Example Application

Given: A reservoir feeding a pipeline through a pump. The elevation change is H. The mass flow rate is $\dot{m}$, and the pump power is $\dot{W}_{\text{pump}}$.
Find: The internal loss of the system.

Assumptions:

- Velocity at the top of the reservoir (1) and the discharge (2) is negligible ($v_1 \approx v_2 \approx 0$).
- Pressures are atmospheric ($p_1 = p_2 = p_0$).
- Fluid is incompressible ($\rho = \text{const}$).

Applying the Extended Bernoulli equation:

$$\left(\frac{1}{2}(0)^2 + gH + \frac{p_0}{\rho}\right) - \left(\frac{1}{2}(0)^2 + g(0) + \frac{p_0}{\rho}\right) = w_{\text{shaft}} - \text{loss} \tag{132}$$

$$gH = w_{\text{shaft}} - \text{loss} \quad \Longrightarrow \quad \text{loss} = \frac{\dot{W}_{\text{pump}}}{\dot{m}} - gH \tag{133}$$

Dimensional consistency check: The units of $w_{\text{shaft}} = \frac{\text{Power}}{\text{Mass Flow}}$ are:

$$\frac{\text{W}}{\text{kg/s}} = \frac{\text{J/s}}{\text{kg/s}} = \text{J/kg} = \frac{\text{N m}}{\text{kg}} = \frac{(\text{kg m/s}^2)\text{m}}{\text{kg}} = \text{m}^2/\text{s}^2 \tag{134}$$

The term gH has units of $(\text{m/s}^2) \cdot \text{m} = \text{m}^2/\text{s}^2$. The units are perfectly consistent.

Expert Analysis: The 2nd Law constraint on 'Loss'

The equation defines loss $= \Delta\tilde{u} - q_{\text{net}}$. The 2nd Law of Thermodynamics in differential form is $Tds = d\tilde{u} + pdv$, or in terms of entropy generation: $ds = \frac{\delta q}{T} + \delta s_{\text{gen}}$.
Combining the 1st and 2nd laws for an incompressible fluid ($dv = 0$), $d\tilde{u} = cdT = \delta q + T\delta s_{\text{gen}}$.
Therefore, the macroscopic loss term is exactly the integral of entropy generated by viscous dissipation:

$$\text{loss} = \int_1^2 T ds_{\text{gen}} \tag{135}$$

Because entropy generation due to friction must be strictly positive ($ds_{\text{gen}} \geq 0$) according to the 2nd Law, we mathematically prove that loss ≥ 0. This forces real fluids to always experience a drop in total mechanical pressure unless external work is applied.

10 Week-10Chapter 6: Differential Analysis

10.1 6.1 Conservation of Mass

10.1.1 6.1.1 From Integral to Differential Form

To establish the differential form of mass conservation, we analyze an arbitrary Control Volume (CV). The total system mass M_{sys} is conserved.

Reynolds Transport Theorem (RTT) for Mass

Applying the RTT to a control volume bounded by a control surface (CS), the rate of change of mass is zero:

$$\frac{DM_{sys}}{Dt} = 0 = \frac{\partial}{\partial t}\int_{CV} \rho \, dV + \int_{CS} \rho\vec{v} \cdot \hat{n} \, dA$$

Application of Gauss's Theorem (Divergence Theorem)

We apply the Divergence Theorem to convert the surface integral of the mass flux into a volume integral:

$$\int_{CS} \rho\vec{v} \cdot \hat{n} \, dA = \int_{CV} \nabla \cdot (\rho\vec{v}) \, dV$$

For a stationary, fixed control volume, the time derivative can be brought inside the integral:

$$\int_{CV} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\vec{v})\right] dV = 0$$

Arbitrary Volume Argument

Because the integral holds true for *any* arbitrary control volume CV, the integrand itself must be identically zero everywhere in the continuum field.

10.1.2 6.1.2 The Continuity Equation

Result: General Forms of Continuity

The fundamental equation for the conservation of mass is given by:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

Applying the product rule ($\nabla \cdot (\rho \vec{v}) = \rho(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla \rho$), we obtain the Eulerian and Lagrangian forms:

- **Eulerian form:** $\dfrac{\partial \rho}{\partial t} + \rho(\nabla \cdot \vec{v}) + \vec{v} \cdot (\nabla \rho) = 0$
- **Lagrangian form:** $\dfrac{D\rho}{Dt} + \rho \nabla \cdot \vec{v} = 0$

Expert Analysis: Incompressible Limit

If the fluid is incompressible, the material derivative of the density is zero ($\frac{D\rho}{Dt} = 0$). The continuity equation rigorously simplifies to a geometric constraint on the velocity field:

$$\nabla \cdot \vec{v} = 0 \quad \text{(Divergence-free field)}$$

10.2 6.2 Newton's 2nd Law (Conservation of Momentum)

10.2.1 6.2.1 Force Analysis and the Stress Tensor

Reynolds Transport Theorem for Momentum

Applying Newton's Second Law ($\frac{D}{Dt}(m\vec{v}) = \sum \vec{F}_{ext}$) using the RTT yields:

$$\frac{\partial}{\partial t} \int_{CV} \rho \vec{v} \, dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \hat{n}) \, dA = \sum \vec{F}_{ext}$$

Forces acting on a fluid element:

- **Body Forces:** Act on the entire volume (e.g., gravity, $\int_{CV} \rho \vec{g} \, dV$).
- **Surface Forces:** Normal pressure forces and viscous stresses acting on boundaries.

Definition: The Stress Tensor

The viscous stress tensor $\overline{\overline{\tau}}$ is defined with components τ_{ij}, where i is the normal to the face and j is the direction of the stress.

$$\overline{\overline{\tau}} = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix}$$

Conservation of angular momentum dictates symmetry: $\tau_{ij} = \tau_{ji}$.

Cauchy Stress Tensor Formulation

The combined total surface stress tensor $\overline{\overline{\sigma}}$ incorporates both thermodynamic pressure p and the deviatoric viscous stress $\overline{\overline{\tau}}$:

$$\overline{\overline{\sigma}} = -p\overline{\overline{I}} + \overline{\overline{\tau}}$$

Thus, surface forces are: $\int_{CS} \overline{\overline{\sigma}} \cdot \hat{n}\, dA = \int_{CV} (-\nabla p + \nabla \cdot \overline{\overline{\tau}})\, dV$.

10.2.2 6.2.2 Cauchy's Equation of Motion

Theorem: Differential Momentum Balance

For an *incompressible* flow ($\rho =$ cte), we analyze the convective flux using the dyadic product:

$$\nabla \cdot (\rho \vec{v} \otimes \vec{v}) = \rho(\vec{v} \cdot \nabla)\vec{v}$$

Substituting this and the force integrals into the RTT formulation leads to Cauchy's Equation.

Cauchy's Equation of Motion

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla p + \rho \vec{g} + \nabla \cdot \overline{\overline{\tau}}$$

This explicitly states that $\rho \vec{a}$ equals the sum of pressure gradients, body forces, and viscous divergence.

10.3 6.3 The Navier-Stokes Equations

10.3.1 6.3.1 Newtonian Constitutive Relation

Definition: Linear Stress-Strain Relationship

For a Newtonian fluid, viscous stresses are linearly proportional to the local strain rate. In 3D tensor notation:

$$\tau_{ij} = \mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$$

where μ is the dynamic viscosity.

10.3.2 6.3.2 Final Derivation for Incompressible Flow

Taking the divergence of the Newtonian viscous stress tensor ($\nabla \cdot \overline{\overline{\tau}}$), we find for the j-th component:

$$\partial_i \tau_{ij} = \mu(\partial_i \partial_i v_j + \partial_i \partial_j v_i) = \mu(\nabla^2 v_j + \partial_j(\nabla \cdot \vec{v}))$$

Since $\nabla \cdot \vec{v} = 0$ for incompressible flow, $\nabla \cdot \overline{\overline{\tau}} = \mu \nabla^2 \vec{v}$.

Result: The Navier-Stokes Equation

Substituting the viscous term into Cauchy's equation yields the governing equation for incompressible flow:

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla p + \rho \vec{g} + \mu \nabla^2 \vec{v}$$

11 Lecture 11: Fluid Dynamics

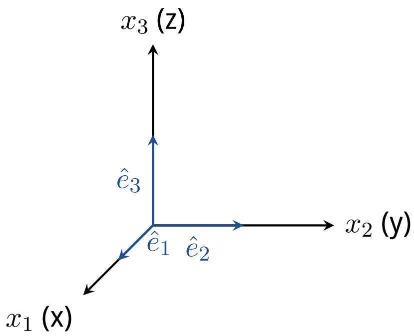
11.1 Intermezzo: Concepts from Analysis III

Reynolds Transport Theorem Notation

$$\int_{CS} \rho\vec{v}(\vec{v}\cdot\hat{n})\,dA + \int_{CV} \frac{\partial}{\partial t}(\rho\vec{v})\,dV = \int_{CV} \rho\left(\frac{\partial\vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right)dV$$

11.1.1 I) Einstein Notation

Cartesian Coordinates:



Cartesian vectors:

$$\vec{v} = v_x\hat{e}_x + v_y\hat{e}_y + v_z\hat{e}_z = v_1\hat{e}_1 + v_2\hat{e}_2 + v_3\hat{e}_3$$

$$\vec{v} = \sum_{i=1}^{3} v_i\hat{e}_i = v_i\hat{e}_i \quad \text{(Einstein Summation Notation)}$$

Derivatives:

$$\frac{\partial}{\partial x_i} = \partial_i$$

$$\partial_i v_i = \nabla\cdot\vec{v}$$

Convective derivative component:

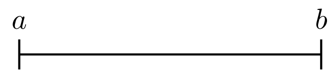
$$u_j\partial_j v_i = u_1\partial_1 v_i + u_2\partial_2 v_i + u_3\partial_3 v_i = \vec{u}\cdot\nabla v_i = (\vec{u}\cdot\nabla\vec{v})_i$$

11.1.2 II) Gauss Theorem / Divergence Theorem

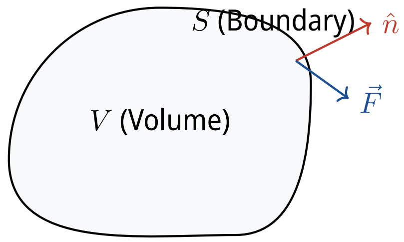
Compare to an $f(x)$ function in 1D:

$$\int_a^b \partial_x f(x)\,dx = f(b) - f(a)$$

Integrating the derivative over the domain = evaluating the function at the boundary.



Generalized for a vector field $\vec{F}$:



Theorem: Divergence Theorem / Gauss

$$\int_V \nabla \cdot \vec{F}\, dV = \int_S (\vec{F} \cdot \hat{n})\, dA = \int_S \vec{F} \cdot d\vec{A}$$

Using Einstein's notation:

$$\int_V \partial_i F_i \, dV = \int_S F_i n_i \, dA$$

Generalized for a Tensor (rank 2) $\overline{\overline{T}}$: Einstein's version:

$$\int_V \partial_i T_{ij}\, dV = \int_S T_{ij} n_i \, dA$$

$$\int_V \nabla \cdot \overline{\overline{T}}\, dV = \int_S \hat{n} \cdot \overline{\overline{T}}\, dA = \int_S \overline{\overline{T}} \cdot \hat{n}\, dA \quad \text{(if } \overline{\overline{T}} \text{ is symmetric)}$$

11.1.3 Outer Products of Vectors

Given $\vec{u}, \vec{v} \in \mathbb{R}^3$ as Cartesian vectors:

$$\vec{u} = u_i \hat{e}_i = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \vec{v} = v_i \hat{e}_i = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Inner Product: $\cdot : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$

$$\vec{v}, \vec{u} \to \vec{v} \cdot \vec{u} = v_i u_i \left(= \sum_{i=1}^{3} v_i u_i \right)$$

Outer Product: $\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^{3\times 3}$

$$\vec{v}, \vec{u} \to \vec{v}\vec{u} \quad \text{(no dot)}$$

$$(\vec{v}\vec{u})_{ij} = v_i u_j$$

$$\vec{v}\vec{u} = \vec{v} \otimes \vec{u} \quad \text{(Continuum Mechanics Notation)}$$

MATLAB-style Notation

Idea: ”Everything is a matrix.”

$$\vec{v} \in \mathbb{R}^{3\times 1}, \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

For matrices, we can define the transpose $\top : \mathbb{R}^{n\times m} \to \mathbb{R}^{m\times n}$. Thus:

$$\vec{v}^\top = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \quad \text{(”row vector” vs ”column vector”)}$$

Matrix Products ($\mathbb{R}^{n\times m} \times \mathbb{R}^{m\times k} \to \mathbb{R}^{n\times k}$):

$$\vec{v}^\top \vec{u} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \vec{v} \cdot \vec{u}$$

$$\vec{v}\vec{u}^\top = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} = \begin{bmatrix} v_1 u_1 & v_1 u_2 & v_1 u_3 \\ v_2 u_1 & v_2 u_2 & v_2 u_3 \\ v_3 u_1 & v_3 u_2 & v_3 u_3 \end{bmatrix} = (\vec{v}\vec{u})_{ij}$$

Where $\vec{v}\vec{u} = \vec{v} \otimes \vec{u}$ (Outer product).

11.2 Navier-Stokes Equations

For an incompressible fluid with $\rho = \text{const}$ and $\mu = \text{const}$:

Navier-Stokes Equations

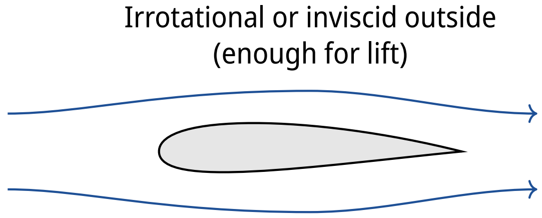
$$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right) = -\nabla p + \rho\vec{g} + \mu\nabla^2\vec{v}$$
$$\nabla\cdot\vec{v} = 0$$

+ Boundary Conditions
+ Initial Conditions

They describe ALL fluid flows.

- Partial differential equations
- 4 equations for 4 unknown fields (v_x, v_y, v_z, p)
- 1st order in time, 2nd order in space
- Non-linear

Outlook: Aim: Describe flows such as flow over an airfoil.



Determine: Lift, Drag, Stall, etc. Problem: Navier-Stokes equations are difficult to solve (impossible analytically for most cases). We look at special cases and approximations:

- Compressible $\leftrightarrow$ Incompressible
- Steady $\leftrightarrow$ Unsteady
- Viscous $\leftrightarrow$ Inviscid
- Rotational $\leftrightarrow$ Irrotational
- 3D $\leftrightarrow$ 2D

11.3 6.3 Reynolds Numbers

Consider the Navier-Stokes momentum equation:

$$\rho\frac{\partial \vec{v}}{\partial t} + \rho\vec{v}\cdot\nabla\vec{v} = -\nabla p + \rho\vec{g} + \mu\nabla^2\vec{v}$$

”Scaling” of the terms:

- Inertial term: $\rho\vec{v}\cdot\nabla\vec{v} \sim \rho\frac{V^2}{L}$
- Viscous term: $\mu\nabla^2\vec{v} \sim \mu\frac{V}{L^2}$

Where V is a characteristic velocity (difference) and L is a characteristic length.

Reynolds Number

$$Re := \frac{\text{Inertial}}{\text{Viscous}} = \frac{\rho\frac{V^2}{L}}{\mu\frac{V}{L^2}} = \frac{\rho V L}{\mu} = \frac{VL}{\nu}$$

where $\nu = \frac{\mu}{\rho}$ is the kinematic viscosity.

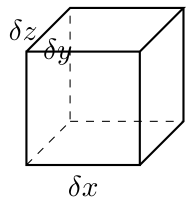
If $Re \gg 1$, we ignore viscosity:

$$\rho\left(\frac{\partial\vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right) = -\nabla p + \rho\vec{g}$$

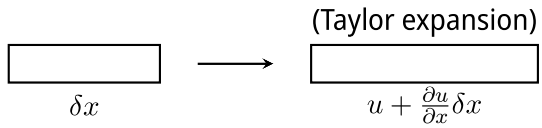
This is the **Euler equation** for inviscid flow.

11.4 6.4 Vorticity and Circulation

11.4.1 Deformation of a fluid element



Deformation $\leftrightarrow$ Linear deformations:



$$\frac{d(\delta x)}{dt} = \frac{\partial u}{\partial x}\delta x \implies \frac{1}{\delta x}\frac{d(\delta x)}{dt} = \frac{\partial u}{\partial x}$$

Also:

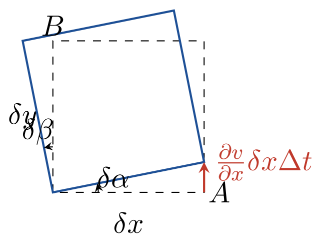
- $\frac{\partial v}{\partial y}$: linear deformation in y
- $\frac{\partial w}{\partial z}$: linear deformation in z

Volume change: $\delta V = \delta x \delta y \delta z$

$$\frac{1}{\delta V}\frac{d(\delta V)}{dt} = \frac{1}{\delta x}\frac{d(\delta x)}{dt} + \frac{1}{\delta y}\frac{d(\delta y)}{dt} + \frac{1}{\delta z}\frac{d(\delta z)}{dt} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \nabla \cdot \vec{v}$$

If $\rho =$ cte (incompressible), then $\nabla \cdot \vec{v} = 0$.

11.4.2 Angular Deformation



Angular velocity of segment OA:

$$\omega_{OA} = \lim_{\Delta t \to 0} \frac{\delta \alpha}{\Delta t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\left(\frac{(\frac{\partial v}{\partial x}\delta x)\Delta t}{\delta x}\right) = \frac{\partial v}{\partial x}$$

Similarly for segment OB (clockwise):

$$\omega_{OB} = -\frac{\partial u}{\partial y}$$

Average rotation (Rotation of the bisector line):

$$\omega_z = \frac{1}{2}(\omega_{OA} + \omega_{OB}) = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

Similarly for the other axes:

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \quad \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

Angular velocity vector:

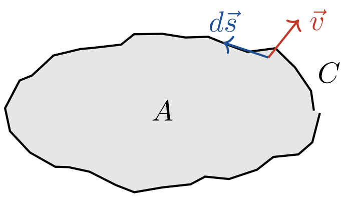
$$\vec{\omega} = \omega_x \hat{\imath} + \omega_y \hat{\jmath} + \omega_z \hat{k} = \frac{1}{2}\nabla \times \vec{v}$$

Definition: Vorticity

The vorticity $\vec{\zeta}$ is defined as twice the angular velocity:

$$\vec{\zeta} = \nabla \times \vec{v} = \text{curl } \vec{v}$$

11.4.3 Circulation



Counter-clockwise path (Right-hand rule)

Vorticity integrated over the Area gives Circulation Γ:

$$\Gamma = \oint_C \vec{v} \cdot d\vec{s} \underset{\text{Stokes}}{=} \int_A (\nabla \times \vec{v}) \cdot d\vec{A}$$

Show: Inviscid flow $\implies$ No change in rotation (Kelvin's Circulation Theorem)

$$\begin{aligned}\frac{D\Gamma}{Dt} = \frac{d\Gamma}{dt} = \frac{d}{dt}\oint_C \vec{v} \cdot d\vec{s} &= \oint_C \frac{d\vec{v}}{dt} \cdot d\vec{s} \\ &= \oint_C \left[\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}\right] \cdot d\vec{s}\end{aligned}$$

Using Navier-Stokes for inviscid flow ($\mu \to 0$):

$$\implies \frac{D\Gamma}{Dt} = \oint_C \frac{1}{\rho}(-\nabla p + \rho\vec{g}) \cdot d\vec{s}$$

Since $\rho\vec{g} = -\rho g\hat{k} = \nabla(-\rho g z)$:

$$= \oint_C \nabla \left(\frac{-p - \rho g z}{\rho} \right) \cdot d\vec{s}$$

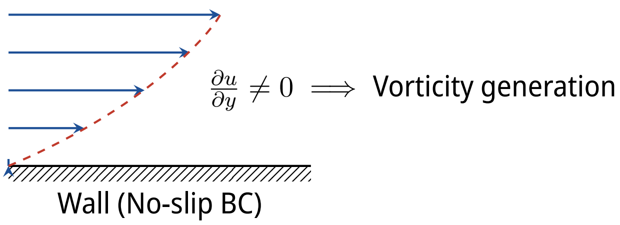
By Stokes' Theorem:

$$\oint_C \nabla\phi \cdot d\vec{s} = \int_A \nabla \times (\nabla\phi) \cdot d\vec{A} = 0 \quad \text{(since } \nabla \times \nabla\phi = \vec{0}\text{)}$$

$$\implies \frac{d\Gamma}{dt} = 0 \quad \text{(for inviscid flow, } \Gamma \text{ is conserved)}$$

Because the curve C is arbitrary, for the vorticity: **An inviscid flow with initially zero vorticity ($\nabla \times \vec{v} = \vec{0}$) remains irrotational.**

Q: How is vorticity generated in a flow? $\implies$ Wall with no-slip boundary condition!



12 Week 12:

12.1 6.5 Irrotational / Potential Flow

Assumptions for Potential Flow

We assume the flow is:

1. **Incompressible** ($\rho = \text{const}$)
2. **Inviscid** ($\mu = 0$)
3. **Irrotational** ($\nabla \times \vec{v} = \vec{0}$)

Since the flow is irrotational ($\nabla \times \vec{v} = \vec{0}$), vector calculus dictates that the velocity field must be the gradient of a scalar field. Therefore, there exists a scalar function Φ such that:

$$\nabla \Phi = \vec{v} \tag{136}$$

where Φ is defined as the **Velocity Potential**.

Using the continuity equation for an incompressible fluid:

$$\nabla \cdot \vec{v} = 0 \quad (\text{since } \rho = \text{const}) \tag{137}$$

Substituting $\vec{v} = \nabla \Phi$ into the continuity equation yields the **Laplace equation** for the velocity potential:

Laplace Equation

$$\nabla^2 \Phi = 0 \tag{138}$$

12.1.1 Momentum Balance (Navier-Stokes for Inviscid, Incompressible Flow)

Starting with the Euler equation (Navier-Stokes assuming inviscid $\mu = 0$ and incompressible $\rho = \text{const}$):

$$\rho \left[\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right] = \rho \vec{g} - \nabla p \tag{139}$$

We utilize the following vector identity for the convective acceleration term:

$$\vec{v} \cdot \nabla \vec{v} = \frac{1}{2} \nabla (\vec{v} \cdot \vec{v}) - \vec{v} \times (\nabla \times \vec{v}) \tag{140}$$

Since the flow is *irrotational*, the vorticity is zero ($\nabla \times \vec{v} = \vec{0}$). The convective acceleration simplifies to:

$$\vec{v} \cdot \nabla \vec{v} = \frac{1}{2} \nabla \|\vec{v}\|^2 = \frac{1}{2} \nabla v^2 \tag{141}$$

Assuming a conservative body force field where $\vec{g} = -g\hat{k}$ (gravity acts downwards), we can write $\rho\vec{g} = -\nabla(\rho g z)$. Substituting this into the momentum balance:

$$\rho\left[\frac{\partial\vec{v}}{\partial t} + \frac{1}{2}\nabla v^2\right] + \nabla p + \nabla(\rho g z) = \vec{0} \tag{142}$$

Now, substituting $\vec{v} = \nabla\Phi$, we can rewrite the unsteady term:

$$\begin{aligned} \rho\frac{\partial}{\partial t}(\nabla\Phi) + \frac{\rho}{2}\nabla v^2 + \nabla p + \nabla(\rho g z) &= \vec{0} \\ \nabla\left[\rho\frac{\partial\Phi}{\partial t} + \frac{\rho}{2}v^2 + p + \rho g z\right] &= \vec{0} \end{aligned}$$

Integrating over the spatial domain gives the **Unsteady Bernoulli Equation**:

Unsteady Bernoulli Equation

$$\rho\frac{\partial\Phi}{\partial t} + \frac{\rho}{2}v^2 + p + \rho g z = \text{constant} \tag{143}$$

If the flow is **steady**, then $\frac{\partial\Phi}{\partial t} = 0$, recovering the classic Bernoulli equation:

$$\frac{\rho}{2}v^2 + p + \rho g z = \text{constant} \tag{144}$$

Solution Strategy for Incompressible, Inviscid, Irrotational Flow

- **Step 1:** Solve the Laplace equation $\nabla^2\Phi = 0$ with appropriate Boundary Conditions (BCs).
- **Step 2:** Compute the velocity field using $\vec{v} = \nabla\Phi$.
- **Step 3:** Use the Bernoulli equation to compute the pressure field p.
- **Step 4:** Integrate the pressure over boundaries to get the net force on a body.

12.2 6.6 2D Flow - Stream Function

From the conservation of mass ($\rho = \text{const}$), we have $\nabla \cdot \vec{v} = 0$. In 2D Cartesian coordinates, this expands to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{145}$$

If a stream function $\Psi(x, y)$ exists with the properties:

$$u = \frac{\partial \Psi}{\partial y}, \quad v = -\frac{\partial \Psi}{\partial x} \tag{146}$$

Let's check continuity:

$$\nabla \cdot \vec{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial \Psi}{\partial y}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial \Psi}{\partial x}\right) = \frac{\partial^2 \Psi}{\partial x \partial y} - \frac{\partial^2 \Psi}{\partial y \partial x} = 0 \tag{147}$$

$\implies$ A 2D velocity field derived from a stream function Ψ will, by construction, automatically satisfy the continuity equation.

12.2.1 Physical Meaning of the Stream Function

Consider the exact differential of Ψ:

$$d\Psi = \frac{\partial \Psi}{\partial x} dx + \frac{\partial \Psi}{\partial y} dy = -v\, dx + u\, dy \tag{148}$$

For a constant value of Ψ (i.e., along a line where $\Psi = \text{const}$), we have $d\Psi = 0$:

$$-v\, dx + u\, dy = 0 \implies v\, dx = u\, dy \implies \frac{v}{u} = \frac{dy}{dx} \tag{149}$$

This is exactly the equation of a **streamline** (where the velocity vector is tangent to the path). Thus, Ψ is constant along a streamline.

Polar Coordinates Extension

For a 2D flow in polar coordinates (r, θ):

$$\vec{v} = v_r \hat{e}_r + v_\theta \hat{e}_\theta$$

The continuity equation is $\nabla \cdot \vec{v} = \frac{1}{r}\frac{\partial}{\partial r}(r v_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} = 0$. The stream function $\Psi(r, \theta)$ is defined as:

$$v_r = \frac{1}{r}\frac{\partial \Psi}{\partial \theta}, \quad v_\theta = -\frac{\partial \Psi}{\partial r} \tag{150}$$

12.3 6.7 2D Potential Flow

Assuming a 2D fluid flow that is **incompressible** and **inviscid**:

- 2D flow $\implies$ There is a stream function Ψ.
- Irrotational $\implies$ There is a velocity potential Φ.

Q: What is the relation between the stream function Ψ and velocity potential Φ?

For Φ, mass conservation requires:

$$0 = \nabla \cdot \vec{v} = \nabla \cdot (\nabla \Phi) = \nabla^2 \Phi \implies \boxed{\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0} \tag{151}$$

For Ψ, the irrotationality condition requires:

$$\vec{0} = \nabla \times \vec{v} \implies 0 = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \tag{152}$$

$$0 = \frac{\partial}{\partial x}\left(-\frac{\partial \Psi}{\partial x}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \Psi}{\partial y}\right) \tag{153}$$

$$0 = -\frac{\partial^2 \Psi}{\partial x^2} - \frac{\partial^2 \Psi}{\partial y^2} \implies \boxed{\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} = 0} \tag{154}$$

Key Result

Both the Velocity Potential Φ and the Stream Function Ψ satisfy the 2D Laplace equation.

12.3.1 Orthogonality of Φ and Ψ

Consider the differential of Φ:

$$d\Phi = \frac{\partial \Phi}{\partial x} dx + \frac{\partial \Phi}{\partial y} dy = u\, dx + v\, dy \tag{155}$$

Along an **equipotential line** ($\Phi =$ const), $d\Phi = 0$:

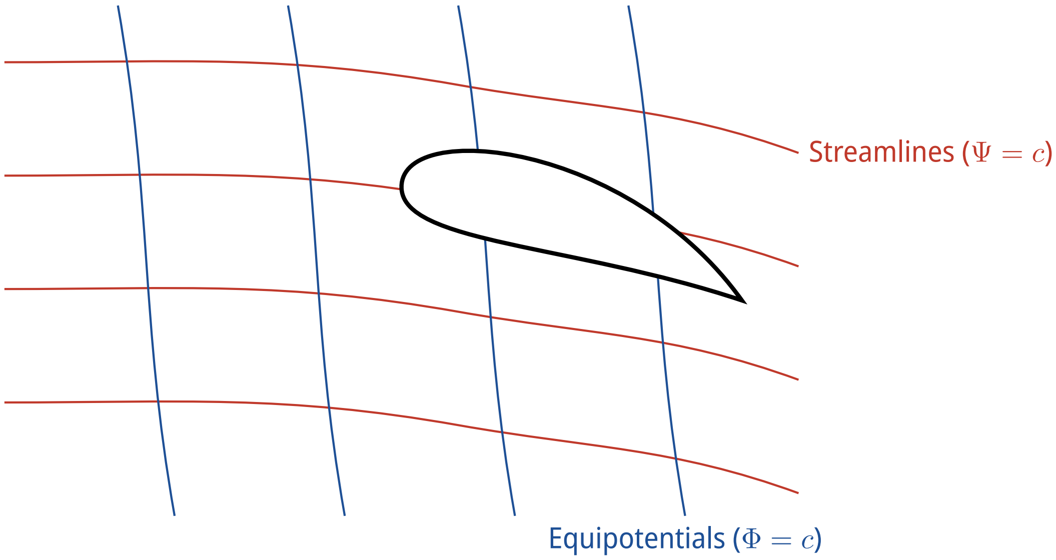
$$u\, dx + v\, dy = 0 \implies \left.\frac{dy}{dx}\right|_{\Phi=\text{const}} = -\frac{u}{v} \tag{156}$$

Recall that for a **streamline** ($\Psi =$ const):

$$\left.\frac{dy}{dx}\right|_{\Psi=\text{const}} = \frac{v}{u} \tag{157}$$

Since the product of their slopes is $\left(-\frac{u}{v}\right) \times \left(\frac{v}{u}\right) = -1$, the equipotential lines ($\Phi =$ const) are **perpendicular** to the streamlines ($\Psi =$ const).

Furthermore, the closer the equipotential lines are, the higher the gradient $\nabla\Phi$, which implies the faster the fluid element is moving.



Example: Stagnation Point Flow

Given the velocity potential:

$$\Phi = \frac{a}{2}(x^2 - y^2)$$

Compute the stream function Ψ (if it exists).
1. Check if incompressible ($\implies \nabla^2\Phi = 0$):

$$\frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} = a + (-a) = 0$$

Since the Laplacian is zero, the flow is incompressible, and a stream function exists.
2. Define Ψ using the Cauchy-Riemann relations: First, derive the velocity field from Φ:

$$u = \frac{\partial\Phi}{\partial x} = ax, \quad v = \frac{\partial\Phi}{\partial y} = -ay$$

Now, equate these to the stream function derivatives:

$$\begin{cases} u = \frac{\partial\Psi}{\partial y} = ax \\ v = -\frac{\partial\Psi}{\partial x} = -ay \end{cases}$$

Integrate the first equation with respect to y:

$$\Psi = axy + f(x)$$

Differentiate this with respect to x and plug into the second equation:

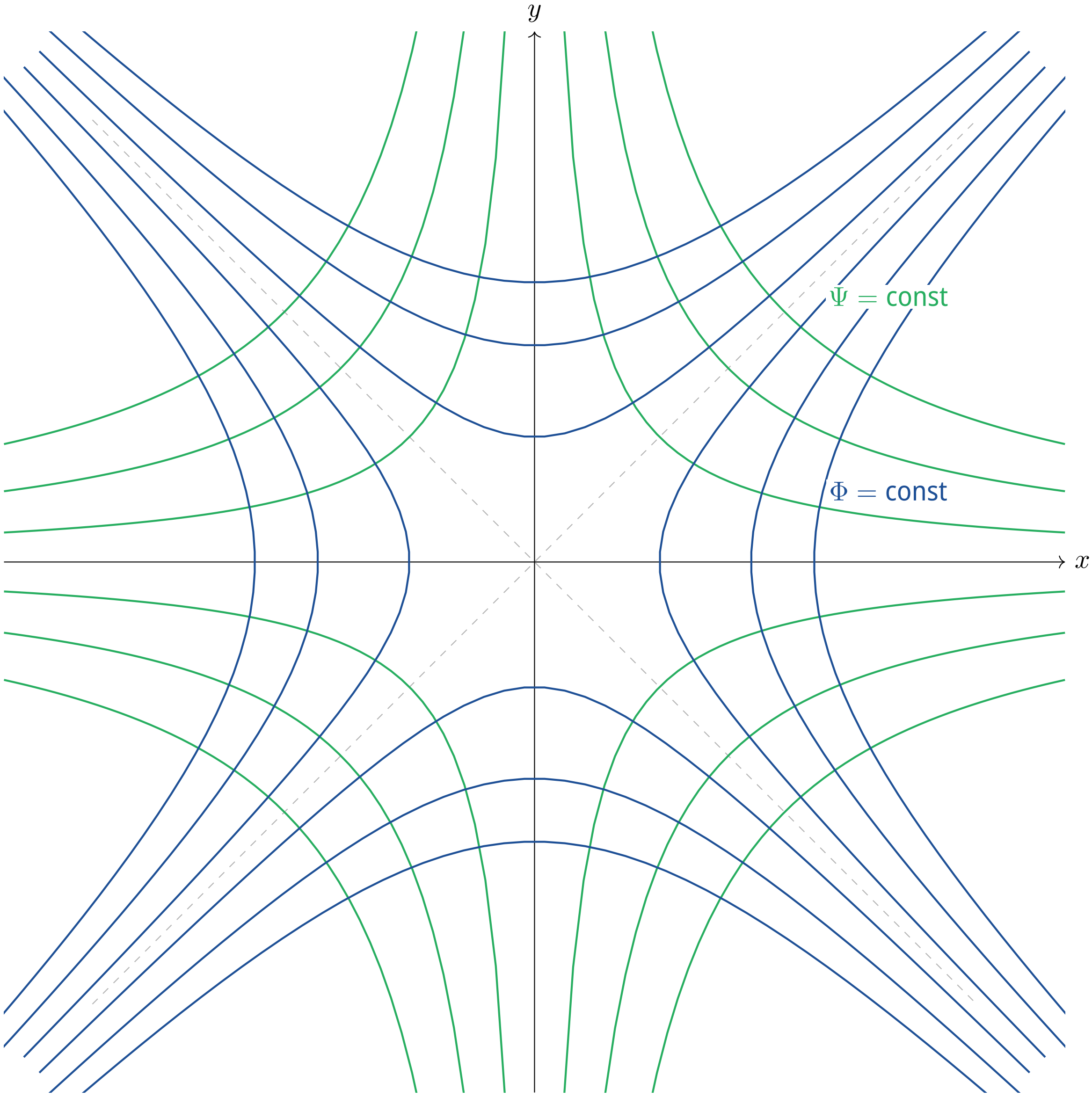
$$\frac{\partial\Psi}{\partial x} = ay + f'(x) = ay \implies f'(x) = 0 \implies f(x) = C$$

Thus, $\Psi = axy + C$. We can set $C = 0$ (free gauge choice).

$$\Psi = axy$$

3. Plot the streamlines and equipotential lines:

- Streamlines: $\Psi =$ const $\implies axy = c \implies y = \frac{c}{ax}$ (Hyperbolas in quadrants $1, 3$ or $2, 4$)
- Equipotentials: $\Phi =$ const $\implies x^2 - y^2 = c'$ (Hyperbolas opening on axes)



Expert Analysis: General Concept for Solving 2D Potential Flow

The foundational principle of 2D potential flow dictates that both the velocity potential and the stream function obey Laplace's equation:

$$\nabla^2\Phi = 0 \quad \text{and} \quad \nabla^2\Psi = 0 \tag{158}$$

Because the Laplace equation is **linear** and **homogeneous**, solutions can be superposed. Complex flow fields around arbitrary bodies can be systematically constructed by linearly combining elementary potential flows (such as uniform flows, sources, sinks, doublets, and point vortices).

13 Week 13 - Potential Flows and Superposition

Due to the linearity of the Laplace equation, complex flow patterns can be constructed by superimposing simple, elementary potential flow solutions.

13.1 6.7.1 Basic Potential Flow Solutions

13.1.1 a) Uniform Flow

Definition: Uniform Flow

A uniform flow is a constant velocity field oriented along a specific direction. For a horizontal flow of magnitude U_0 along the $+x$ axis, the velocity components in Cartesian coordinates are:

$$u = U_0 = \frac{\partial \Phi}{\partial x} = \frac{\partial \Psi}{\partial y}$$

$$v = 0 = \frac{\partial \Phi}{\partial y} = -\frac{\partial \Psi}{\partial x}$$

Integrating these relations yields the potentials for a uniform flow:

Uniform Flow Potentials

$$\begin{aligned} \text{Velocity Potential:} \quad & \Phi(x, y) = U_0 x \\ \text{Stream Function:} \quad & \Psi(x, y) = U_0 y \end{aligned}$$

In polar coordinates ($x = r\cos\theta,\ y = r\sin\theta$), these potentials are expressed as:

$$\Phi(r, \theta) = U_0 r \cos\theta \qquad \Psi(r, \theta) = U_0 r \sin\theta$$

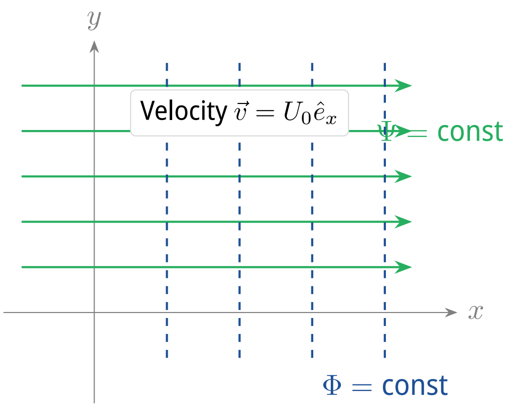


Figure 1: Equipotential lines (Φ, dashed vertical) and streamlines (Ψ, solid horizontal) for a uniform flow parallel to the x-axis.

13.1.2 b) Source / Sink (Line Source)

Let m represent the volumetric rate of flow emanating from a line source per unit length perpendicular to the plane of the flow. If $m > 0$, fluid flows radially outwards (source). If $m < 0$, fluid flows radially inwards (sink).

Using polar coordinates with velocity field $\vec{v} = v_r \hat{e}_r + v_\theta \hat{e}_\theta$, we apply the conservation of mass through a control surface of radius r:

$$m\,\Delta t = (2\pi r \cdot 1) v_r\,\Delta t \implies v_r = \frac{m}{2\pi r}, \quad v_\theta = 0$$

Line Source Potentials

$$\text{Radial Velocity:} \quad v_r = \frac{m}{2\pi r} = \frac{\partial \Phi}{\partial r} = \frac{1}{r}\frac{\partial \Psi}{\partial \theta}$$

$$\text{Azimuthal Velocity:} \quad v_\theta = 0 = \frac{1}{r}\frac{\partial \Phi}{\partial \theta} = -\frac{\partial \Psi}{\partial r}$$

Integrating the equations in the box yields:

$$\Phi(r,\theta) = \frac{m}{2\pi}\ln(r) \qquad \Psi(r,\theta) = \frac{m}{2\pi}\theta$$

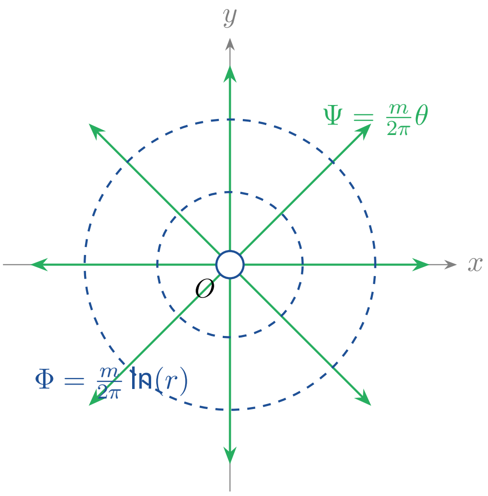


Figure 2: Streamlines (radial) and equipotential lines (concentric circles) for a line source at the origin.

13.1.3 c) The Line Vortex (Free / Irrotational Vortex)

A free vortex flow features concentric circular streamlines with a velocity potential that is mathematically conjugate to that of a source flow. By swapping the functional forms of the potential and stream functions, we obtain:

Free Vortex Potentials

$$\Phi(r,\theta) = K\theta$$
$$\Psi(r,\theta) = -K\ln(r)$$

Where K is a constant representing the strength of the vortex. The velocity components are:

$$v_r = \frac{\partial\Phi}{\partial r} = 0 = \frac{1}{r}\frac{\partial\Psi}{\partial\theta}$$
$$v_\theta = \frac{1}{r}\frac{\partial\Phi}{\partial\theta} = \frac{K}{r} = -\frac{\partial\Psi}{\partial r}$$

In a free vortex, the velocity is inversely proportional to the radius ($v_\theta \propto 1/r$).

13.1.4 d) Expert Analysis: Irrotationality and Circulation of a Free Vortex

Expert Analysis: How can a vortex be irrotational?

It may seem counter-intuitive that a flow spinning in circles is mathematically classified as "irrotational" ($\nabla \times \vec{v} = 0$). To clarify this, we distinguish between a **forced (rotational) vortex** and a **free (irrotational) vortex**:

- **Forced Vortex (Solid-body rotation):** $v_\theta = \omega r$. Fluid elements rotate about their own axes. Vorticity is non-zero everywhere ($\vec{\omega} = \nabla \times \vec{v} = 2\omega \hat{e}_z \neq 0$).
- **Free Vortex (Potential flow):** $v_\theta = K/r$. Local fluid elements translate along a circular path but do not rotate about their own centers. Calculating the vorticity in polar coordinates:

$$\omega_z = (\nabla \times \vec{v})_z = \frac{1}{r}\left[\frac{\partial(r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta}\right] = \frac{1}{r}\left[\frac{\partial(K)}{\partial r} - 0\right] = 0 \quad (\text{for } r \neq 0)$$

Let us evaluate the circulation Γ around a circular contour of radius R centered at the origin:

$$\Gamma = \oint \vec{v} \cdot d\vec{s} = \int_0^{2\pi} (v_\theta \hat{e}_\theta) \cdot (R\, d\theta\, \hat{e}_\theta) = \int_0^{2\pi} \frac{K}{R} R\, d\theta = 2\pi K$$

Thus, the vortex strength constant K is directly related to the circulation Γ:

$$K = \frac{\Gamma}{2\pi}$$

Any contour enclosing the origin yields $\Gamma \neq 0$, indicating a line singularity at $r = 0$. However, for any closed loop that does not contain the origin, Stokes' theorem holds with $\oint \vec{v} \cdot d\vec{s} = \iint (\nabla \times \vec{v}) \cdot d\vec{A} = 0$, confirming the flow field is irrotational everywhere except at the singular origin point.

The final potential and stream functions for an irrotational vortex of circulation Γ are:

$$\Phi = \frac{\Gamma}{2\pi}\theta \qquad \Psi = -\frac{\Gamma}{2\pi}\ln(r)$$

13.1.5 e) Doublet

A doublet is formed by placing a line source of strength $+m$ and a sink of strength $-m$ at a distance a apart, and taking the limit as $a \to 0$ while keeping the product $m \cdot a$ constant.

The stream function of the combined system is:

$$\Psi = \frac{m}{2\pi}(\theta_2 - \theta_1) = -\frac{m}{2\pi}\Delta\theta$$

For small separation distance a, we observe that $\Delta\theta \approx \frac{a \sin\theta}{r}$. Taking the limit as $a \to 0$ and $m \to \infty$ while holding the doublet strength $H = \frac{m \cdot a}{\pi}$ constant:

$$\Psi = \lim_{a \to 0}\left(-\frac{ma\sin\theta}{2\pi r}\right) = -\frac{H\sin\theta}{r} \quad \text{where} \quad H = \frac{ma}{2\pi}$$

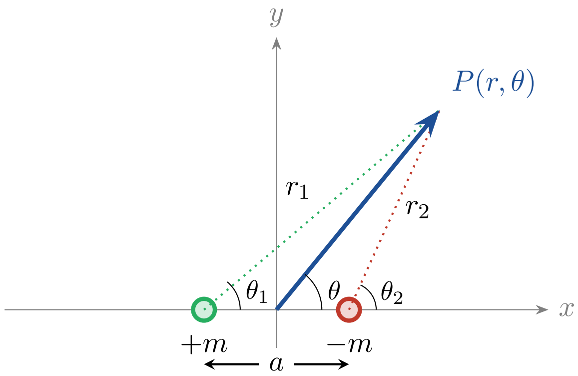


Figure 3: Geometric construction of a doublet from a source ($+m$) and a sink ($-m$).

Doublet Potentials

$$\Phi(r,\theta) = \frac{H\cos\theta}{r}$$
$$\Psi(r,\theta) = -\frac{H\sin\theta}{r}$$

The corresponding polar velocity components are:

$$v_r = \frac{1}{r}\frac{\partial\Psi}{\partial\theta} = -\frac{H\cos\theta}{r^2}$$
$$v_\theta = -\frac{\partial\Psi}{\partial r} = -\frac{H\sin\theta}{r^2}$$

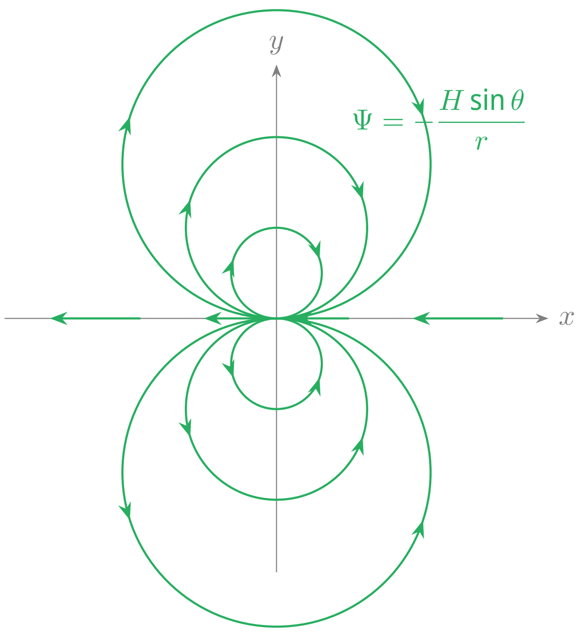


Figure 4: Doublet streamlines forming circular loops tangent to the x-axis at the origin.

13.2 6.7.2 Superpositions of Basic Solutions

13.2.1 Source in a Uniform Flow: The Rankine Half-Body

By combining a uniform flow parallel to the x-axis and a line source located at the origin, we construct the flow field around a semi-infinite body known as a Rankine half-body.

Rankine Half-Body Potentials

$$\Psi = \Psi_{\text{UF}} + \Psi_{\text{Source}} = U_0 r \sin\theta + \frac{m}{2\pi}\theta$$
$$\Phi = \Phi_{\text{UF}} + \Phi_{\text{Source}} = U_0 r \cos\theta + \frac{m}{2\pi}\ln(r)$$

The velocity components are:

$$v_r = \frac{1}{r}\frac{\partial\Psi}{\partial\theta} = U_0\cos\theta + \frac{m}{2\pi r}$$
$$v_\theta = -\frac{\partial\Psi}{\partial r} = -U_0\sin\theta$$

Finding the Stagnation Point To find the stagnation point, we set both velocity components to zero:

$$v_\theta = -U_0\sin\theta = 0 \implies \theta = \pi \quad \text{(upstream on the negative } x\text{-axis)}$$
$$v_r\big|_{(r_{sp},\pi)} = -U_0 + \frac{m}{2\pi r_{sp}} = 0 \implies r_{sp} = \frac{m}{2\pi U_0}$$

Thus, the stagnation point is located at polar coordinates $(r,\theta) = \left(\frac{m}{2\pi U_0}, \pi\right)$, which corresponds to Cartesian coordinates $(x,y) = \left(-\frac{m}{2\pi U_0}, 0\right)$.

The Stagnation Streamline and Body Shape Evaluating the stream function Ψ at the stagnation point yields:

$$\Psi_{\text{stagnation}} = \Psi\left(\frac{m}{2\pi U_0}, \pi\right) = U_0\left(\frac{m}{2\pi U_0}\right)\sin(\pi) + \frac{m}{2\pi}\pi = \frac{m}{2}$$

The equation of the dividing stagnation streamline $\Psi = \frac{m}{2}$ is given by:

$$U_0 r\sin\theta + \frac{m}{2\pi}\theta = \frac{m}{2} \implies r = \frac{m}{2\pi U_0}\frac{\pi-\theta}{\sin\theta}$$

Analyzing the key asymptotic dimensions of this body geometry:

- **At $\theta = \frac{\pi}{2}$ (directly above the source):**

$$r_{\theta=\pi/2} = \frac{m}{2\pi U_0}\frac{\pi - \pi/2}{\sin(\pi/2)} = \frac{m}{4U_0}$$

  Thus, the half-width y at this position is $y = r\sin(\pi/2) = \frac{m}{4U_0}$.

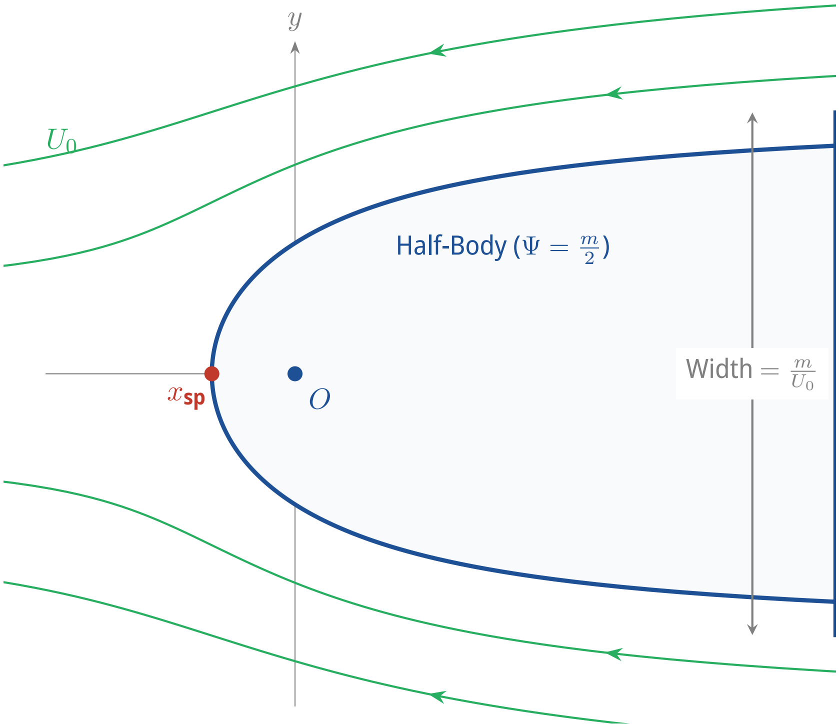


Figure 5: Potential flow streamlines around a Rankine half-body.

- **As $\theta \to 0$ (far downstream):**

$$y_\infty = \lim_{\theta \to 0^+} (r \sin\theta) = \lim_{\theta \to 0^+} \left(\frac{m(\pi - \theta)}{2\pi U_0} \right) = \frac{m}{2U_0}$$

Thus, the total asymptotic height of the half-body is $2y_\infty = \frac{m}{U_0}$.

Rankine Half-Body Wind Tunnel Analysis

Problem Statement:
A model representing a bridge pier is designed as a Rankine half-body. The wind tunnel air speed is $U_0 = 64$ km/h. The total asymptotic width of the half-body downstream is $2y_\infty = 60$ m.

1. Compute the source strength m required to model this flow.
2. Calculate the wind velocity magnitude at the top vertex point of the body where $\theta = \pi/2$.
3. Find the static pressure change between the free-stream flow and this vertex point.

Solution:

1. First, we convert the wind velocity to SI units:

$$U_0 = 64 \text{ km/h} = \frac{64 \times 1000}{3600} \text{ m/s} \approx 17.78 \text{ m/s}$$

Given the total asymptotic width is $2y_\infty = 60$ m, the half-width is $y_\infty = 30$ m. Using the asymptotic width relation:

$$y_\infty = \frac{m}{2U_0} \implies m = 2U_0 y_\infty = 2 \times 17.78 \text{ m/s} \times 30 \text{ m} \approx 1066.7 \text{ m}^2/\text{s}$$

2. At $\theta = \pi/2$, the radial coordinate along the surface of the body is:

$$r = \frac{m}{4U_0} = \frac{1066.7}{4 \times 17.78} = 15 \text{ m}$$

We evaluate the velocity components at $(r, \theta) = (15 \text{ m}, \pi/2)$:

$$v_r = U_0 \cos(\pi/2) + \frac{m}{2\pi r} = 0 + \frac{1066.7}{2\pi \times 15} = \frac{2U_0}{\pi} \approx 11.32 \text{ m/s}$$

$$v_\theta = -U_0 \sin(\pi/2) = -U_0 \approx -17.78 \text{ m/s}$$

The total velocity magnitude squared is:

$$V^2 = v_r^2 + v_\theta^2 = \left(\frac{2U_0}{\pi}\right)^2 + (-U_0)^2 = U_0^2 \left(1 + \frac{4}{\pi^2}\right)$$

Taking the square root:

$$V = U_0 \sqrt{1 + \frac{4}{\pi^2}} \approx 17.78 \times 1.1854 \approx 21.08 \text{ m/s} \approx 76 \text{ km/h}$$

3. Using Bernoulli's equation, assuming horizontal flow (negligible gravity difference):

$$p_\infty + \frac{1}{2}\rho U_0^2 = p_{\text{vertex}} + \frac{1}{2}\rho V^2$$

$$p_{\text{vertex}} - p_\infty = \frac{1}{2}\rho \left(U_0^2 - V^2\right) = \frac{1}{2}\rho U_0^2 \left[1 - \left(1 + \frac{4}{\pi^2}\right)\right] = -\frac{2\rho U_0^2}{\pi^2}$$

Taking standard air density $\rho \approx 1.2$ kg/m^3:

$$\Delta p = p_{\text{vertex}} - p_\infty = -\frac{2 \times 1.2 \times (17.78)^2}{\pi^2} \approx -76.8 \text{ Pa}$$

There is a drop in static pressure of approximately 76.8 Pa due to flow acceleration.

Rankine Ovals

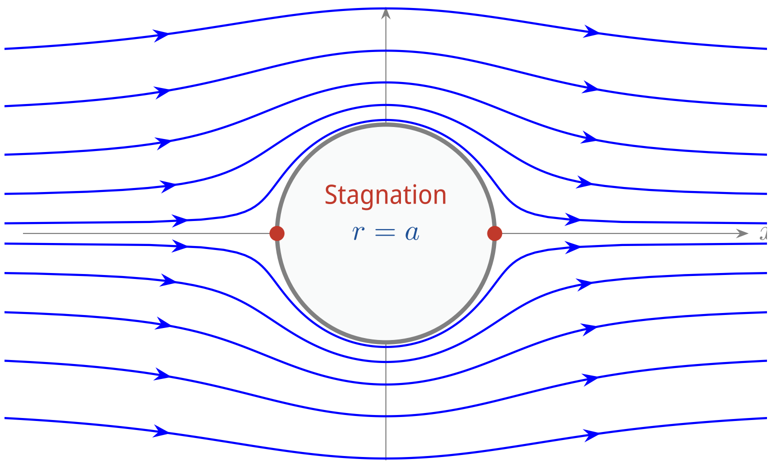
By superimposing a uniform flow, a line source ($+m$), and a line sink ($-m$) of equal strength separated by a distance $2a$ along the x-axis, we generate closed oval-shaped bodies known as **Rankine Ovals**.

13.2.2 Flow Around a Circular Cylinder (Uniform Flow + Doublet)

Superimposing a uniform flow and a doublet at the origin creates the flow pattern around a solid circular cylinder.

Cylinder Flow Potentials

$$\Psi = U_0 \left(1 - \frac{H}{U_0 r^2}\right) r \sin\theta$$
$$\Phi = U_0 \left(1 + \frac{H}{U_0 r^2}\right) r \cos\theta$$



The stream function $\Psi = 0$ must represent the boundary of a solid cylinder of radius $r = a$. Setting $\Psi = 0$ at $r = a$:

$$U_0 \left(1 - \frac{H}{U_0 a^2}\right) a \sin\theta = 0 \implies H = U_0 a^2$$

Substituting the doublet strength $H = U_0 a^2$ back into the potentials yields:

Final Cylinder Potentials

$$\Psi = U_0 \left(1 - \frac{a^2}{r^2}\right) r \sin\theta$$
$$\Phi = U_0 \left(1 + \frac{a^2}{r^2}\right) r \cos\theta$$

The velocity components are:

$$v_r = \frac{\partial \Phi}{\partial r} = U_0 \left(1 - \frac{a^2}{r^2}\right) \cos\theta$$

$$v_\theta = \frac{1}{r}\frac{\partial \Phi}{\partial \theta} = -U_0 \left(1 + \frac{a^2}{r^2}\right) \sin\theta$$

On the Surface of the Cylinder ($r = a$) Evaluating the velocities on the cylinder boundary:

$$v_r\big|_{r=a} = 0$$

$$v_\theta\big|_{r=a} = -2U_0 \sin\theta$$

- The radial velocity is zero, satisfying the solid wall boundary condition.
- Stagnation points occur where $v_\theta = 0 \implies \theta = 0$ and $\theta = \pi$.
- The maximum velocity magnitude is $|v_\theta|_{\text{max}} = 2U_0$, occurring at the top and bottom of the cylinder ($\theta = \pi/2$ and $\theta = 3\pi/2$).

Pressure Distribution and the Pressure Coefficient Applying Bernoulli's equation between the upstream free-stream (p_∞, velocity U_0) and a point on the surface of the cylinder (p_s, velocity $v_\theta = -2U_0 \sin\theta$):

$$p_\infty + \frac{1}{2}\rho U_0^2 = p_s + \frac{1}{2}\rho v_\theta^2$$

$$p_s = p_\infty + \frac{1}{2}\rho U_0^2 \left(1 - \frac{v_\theta^2}{U_0^2}\right) = p_\infty + \frac{1}{2}\rho U_0^2 \left(1 - 4\sin^2\theta\right)$$

We define the dimensionless pressure coefficient, C_p:

Surface Pressure Coefficient

$$C_p = \frac{p_s - p_\infty}{\frac{1}{2}\rho U_0^2} = 1 - 4\sin^2\theta$$

Let us analyze the key coordinates of this pressure distribution:

- At stagnation points ($\theta = 0, \pi$): $C_p = 1$ (maximum pressure).
- At the sides ($\theta = \pi/2, 3\pi/2$): $C_p = -3$ (minimum pressure).
- Zero-crossing points ($C_p = 0 \implies \sin\theta = \pm 1/2$): $\theta = \pm 30°, \pm 150°$.

13.3 6.7.3 Aerodynamic Forces: D'Alembert's Paradox

The net aerodynamic forces per unit depth acting on the cylinder are calculated by integrating the surface pressure distribution. Let a be the radius of the cylinder.

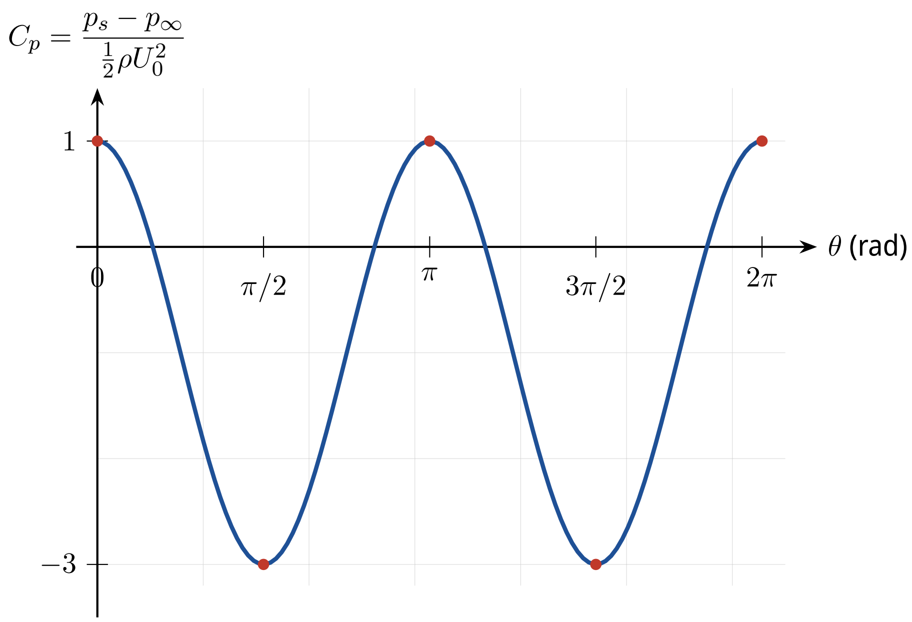


Figure 6: Dimensionless pressure coefficient distribution C_p along the cylinder boundary.

13.3.1 Lift Force (Vertical Component)

$$F_L = -\int_0^{2\pi} p_s \sin\theta \, a \, d\theta$$

Substituting $p_s(\theta) = p_\infty + \frac{1}{2}\rho U_0^2(1 - 4\sin^2\theta)$:

$$F_L = -a\int_0^{2\pi} \left[p_\infty + \frac{1}{2}\rho U_0^2(1 - 4\sin^2\theta)\right] \sin\theta \, d\theta$$

Due to top-bottom symmetry, the integral of any odd power of $\sin\theta$ over $[0, 2\pi]$ is zero:

$$\int_0^{2\pi} \sin\theta \, d\theta = 0, \qquad \int_0^{2\pi} \sin^3\theta \, d\theta = 0 \implies F_L = 0$$

13.3.2 Drag Force (Horizontal Component)

$$F_D = -\int_0^{2\pi} p_s \cos\theta \, a \, d\theta$$

$$F_D = -a\int_0^{2\pi} \left[p_\infty + \frac{1}{2}\rho U_0^2(1 - 4\sin^2\theta)\right] \cos\theta \, d\theta$$

Evaluating the terms:

$$\int_0^{2\pi} \cos\theta \, d\theta = 0$$

$$\int_0^{2\pi} \sin^2\theta \cos\theta \, d\theta = \left[\frac{\sin^3\theta}{3}\right]_0^{2\pi} = 0 \implies F_D = 0$$

D’Alembert’s Paradox (1752)

The prediction of zero drag force for a body moving through an incompressible, inviscid, and irrotational fluid is known as **D’Alembert’s Paradox**.
This prediction directly contradicts physical observation, where all real fluids exert a drag force. This paradox was resolved in 1904 by Ludwig Prandtl with the introduction of the **Boundary Layer Theory**. For high Reynolds number flows of fluids with small but non-zero viscosity, viscous effects are concentrated within a thin boundary layer near the solid wall. This layer experiences flow separation, leading to a low-pressure wake behind the body, which generates pressure drag.

13.4 6.7.4 Expert Analysis: Can We Create Lift?

Expert Analysis: Generating Aerodynamic Lift via Circulation

Since a symmetric, non-rotating cylinder in a uniform flow yields zero lift and drag, we ask: *How can we modify this system to generate lift?*
We can achieve this by superimposing a **free vortex** of circulation Γ centered at the origin onto the cylinder flow. The combined stream function becomes:

$$\Psi = U_0 \left(1 - \frac{a^2}{r^2}\right) r \sin\theta - \frac{\Gamma}{2\pi} \ln\left(\frac{r}{a}\right)$$

On the cylinder surface $r = a$, the radial velocity remains $v_r = 0$, but the azimuthal velocity is:

$$v_\theta\Big|_{r=a} = -2U_0 \sin\theta - \frac{\Gamma}{2\pi a}$$

This asymmetry in velocity speeds up the flow on the top of the cylinder (where $\sin\theta > 0$, and the vortex velocity adds to the uniform flow) and slows it down on the bottom. According to Bernoulli's equation, this velocity difference results in lower pressure on top and higher pressure on the bottom, generating a net upward lift force.
Integrating the asymmetric surface pressure:

$$F_L = -\int_0^{2\pi} p_s \sin\theta \, a \, d\theta = -\rho U_0 \Gamma$$

This fundamental relation is known as the **Kutta-Joukowski theorem**. It demonstrates that circulation is the physical mechanism behind the generation of lift. This principle explains the Magnus effect (the lateral force on spinning sports balls) and governs lift generation on complex aerodynamic airfoils.

14 Week 14: Potential and Viscous Flows in Fluid Mechanics

14.1 Flow Around a Rotating Cylinder (The Magnus Effect)

Building upon the ideal potential flow theory, we consider the superposition of a uniform, inviscid, irrotational flow past a cylinder of radius a and a free line vortex centered at the origin. This classical model demonstrates how circulation around a body generates a lifting force.

Superposition of Potential Flows

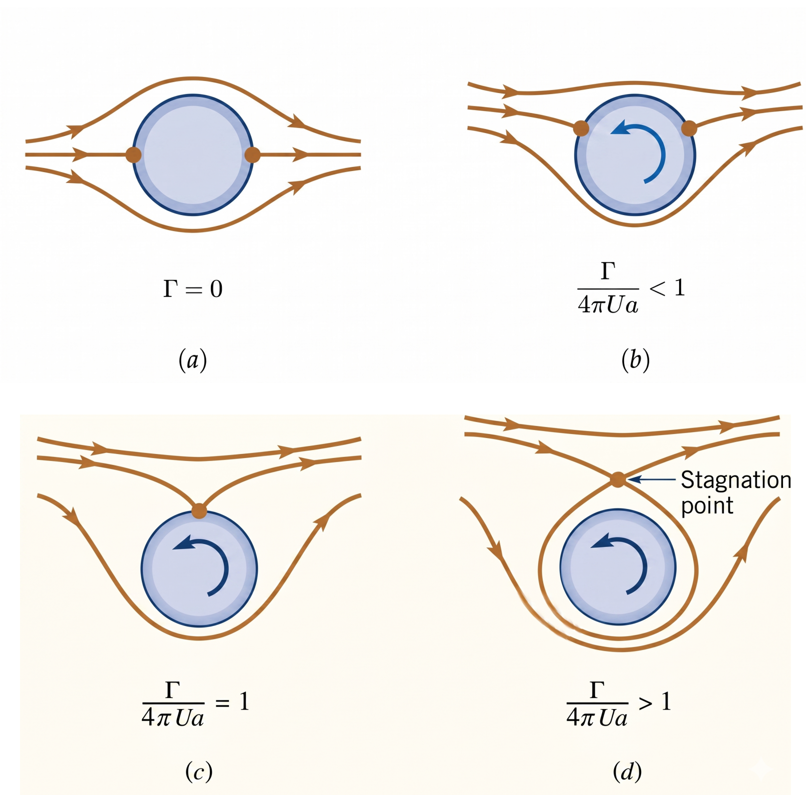
The total flow field is modeled by superposing the stream function Ψ and velocity potential ϕ of a uniform flow around a cylinder with those of a free vortex:

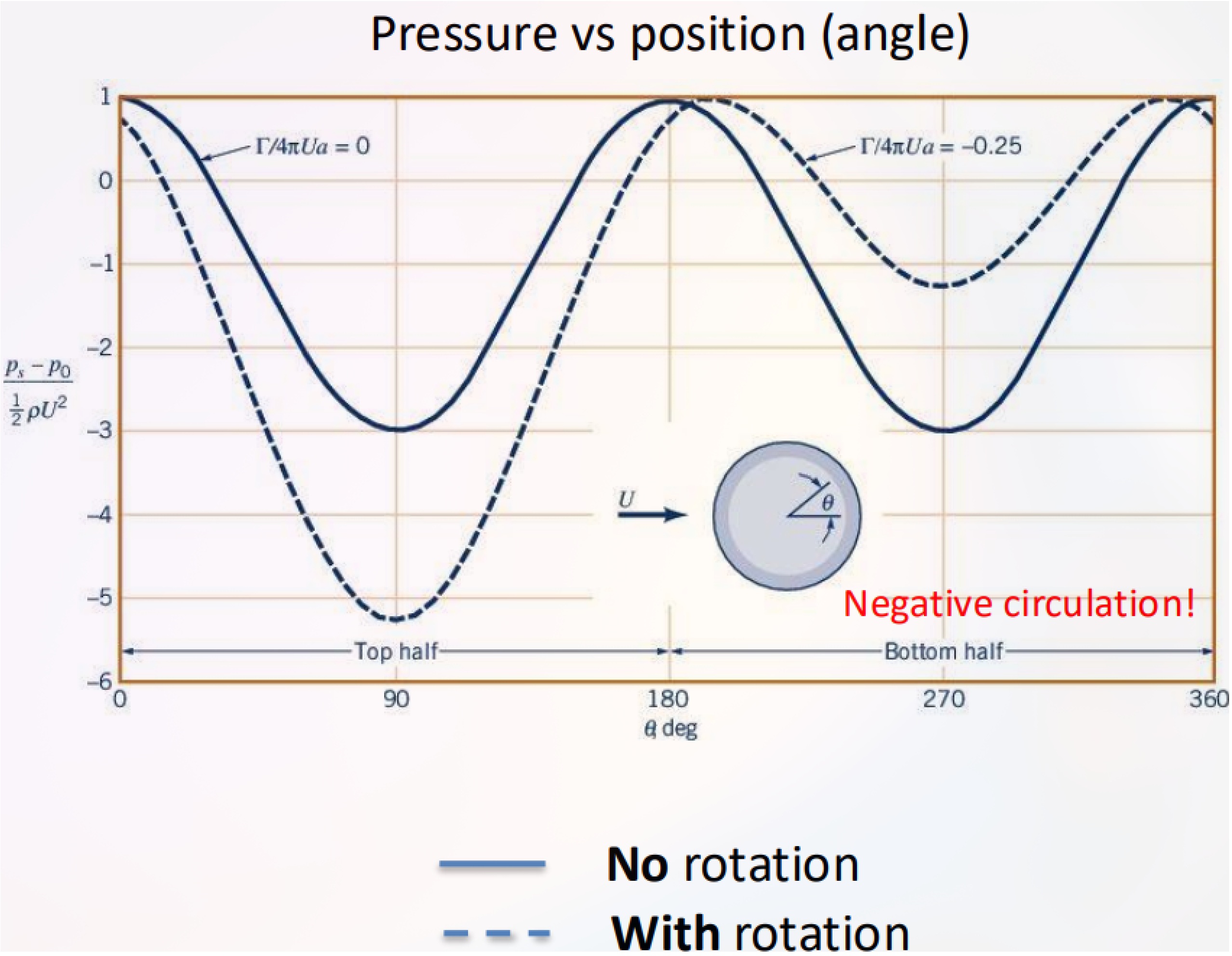
$$\Psi(r,\theta) = \Psi_{\text{cyl}} + \Psi_{\text{vortex}} = U_0 \left(r - \frac{a^2}{r} \right) \sin\theta - \frac{\Gamma}{2\pi} \ln\left(\frac{r}{a}\right) \tag{159}$$

$$\phi(r,\theta) = \phi_{\text{cyl}} + \phi_{\text{vortex}} = U_0 \left(r + \frac{a^2}{r} \right) \cos\theta + \frac{\Gamma}{2\pi}\theta \tag{160}$$

where:

- U_0 is the uniform free-stream velocity pointing in the $+x$-direction ($\theta = 0$).
- a is the radius of the cylinder.
- Γ is the circulation of the free vortex (defined positive in the counter-clockwise direction).





14.2 Pressure Distribution and Force Derivation

The surface pressure $p(\theta)$ is obtained from Bernoulli's equation for steady, incompressible, irrotational flow:

$$p(\theta) + \frac{1}{2}\rho v^2\big|_{r=a} = p_\infty + \frac{1}{2}\rho U_0^2 \tag{161}$$

Substituting $v^2\big|_{r=a} = v_\theta^2\big|_{r=a}$, we expand the pressure field:

$$p(\theta) = p_\infty + \frac{1}{2}\rho U_0^2 - \frac{1}{2}\rho\left(-2U_0 \sin\theta + \frac{\Gamma}{2\pi a}\right)^2 \tag{162}$$

Surface Pressure Distribution

$$p(\theta) = p_\infty + \frac{1}{2}\rho U_0^2\left(1 - 4\sin^2\theta + \frac{2\Gamma\sin\theta}{\pi a U_0} - \frac{\Gamma^2}{4\pi^2 a^2 U_0^2}\right) \tag{163}$$

To find the net aerodynamic forces per unit depth acting on the cylinder, we integrate the surface pressure:

$$F_x = -\int_0^{2\pi} p(\theta)\cos\theta\, a\, d\theta, \quad F_y = -\int_0^{2\pi} p(\theta)\sin\theta\, a\, d\theta \tag{164}$$

1. Drag Force (F_x): Due to the symmetry of the pressure profile under the transformation $x \to -x$ (which corresponds to $\theta \to \pi - \theta$):

$$p(\theta) = p(\pi - \theta) \quad \text{while} \quad \cos(\pi - \theta) = -\cos\theta \tag{165}$$

Integrating these terms yields:

$$F_x = 0 \quad \text{(D'Alembert's Paradox)} \tag{166}$$

2. Lift Force (F_y): Evaluating the integral for the vertical force component, we observe that the only non-vanishing contribution comes from the term proportional to $\sin^2\theta$ after multiplying by $\sin\theta$:

$$F_y = -\int_0^{2\pi} \left[\frac{1}{2}\rho U_0^2 \left(\frac{2\Gamma \sin\theta}{\pi a U_0}\right)\right] \sin\theta \, a \, d\theta \tag{167}$$

Simplifying the constants and evaluating the standard integral $\int_0^{2\pi} \sin^2\theta \, d\theta = \pi$:

$$F_y = -\frac{\rho U_0 \Gamma}{\pi} \int_0^{2\pi} \sin^2\theta \, d\theta = -\rho U_0 \Gamma \tag{168}$$

Result: The Kutta-Joukowski Lift Theorem

The lift force per unit span is perpendicular to the free-stream direction and depends uniquely on the fluid density, the free-stream velocity, and the circulation:

$$F_{\text{Lift}} = F_y = -\rho U_0 \Gamma \tag{169}$$

14.3 Lift on an Aircraft Wing

For a 2D infinite wing operating at high Reynolds number ($Re \gg 1$), the flow outside the boundary layer is modeled as an inviscid, irrotational potential flow.

Potential Flow Modeling of an Airfoil

The general aerodynamic results for an airfoil parallel the cylinder case:

- **Drag:** $F_{\text{Drag}} = 0$ (D'Alembert's Paradox for 2D inviscid flow).
- **Lift:** $F_{\text{Lift}} = -\rho U_0 \Gamma$.

The circulation is calculated along any closed contour C enclosing the airfoil:

$$\Gamma = \oint_C \vec{v} \cdot d\vec{s} \tag{170}$$

Generation of Circulation and the Kutta Condition

In an ideal inviscid fluid, any value of circulation Γ is a valid mathematical solution. In reality, viscous effects at the sharp trailing edge of the airfoil dictate a unique circulation. The flow must leave the trailing edge smoothly without physically impossible infinite velocities. This physical requirement is known as the **Kutta Condition**, which determines the precise circulation Γ needed to generate realistic lift.

14.4 Viscous Flow and the Navier-Stokes Equations

When accounting for viscosity, we must transition from the Euler equations to the Navier-Stokes equations. For an incompressible Newtonian fluid, the conservation of momentum and mass are governed by:

Incompressible Navier-Stokes Equations

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right) = -\nabla p + \rho\vec{g} + \mu\nabla^2\vec{v} \tag{171}$$

$$\nabla\cdot\vec{v} = 0 \tag{172}$$

where μ is the dynamic viscosity and $\vec{g}$ represents the gravitational body force.

The No-Slip Boundary Condition

Viscosity introduces friction at solid boundaries. As a consequence, the fluid velocity must exactly match the velocity of the solid boundary:

$$\vec{v}\big|_{\text{wall}} = \vec{v}_{\text{wall}} \tag{173}$$

This condition applies to both the normal component (no penetration) and the tangential component (no slip) of the velocity vector.

The presence of the non-linear advective term $\vec{v}\cdot\nabla\vec{v}$ makes the Navier-Stokes equations notoriously difficult to solve analytically. However, for specific symmetric geometries, this non-linear term vanishes identically, allowing for exact analytical solutions.

Hydrodynamic Stability Constraint

While the exact mathematical solutions derived under the parallel flow assumption ($\vec{v}\cdot\nabla\vec{v} = 0$) hold true across all parameter spaces algebraically, they are only physically realized at low to moderate Reynolds numbers. High-velocity conditions subject the laminar base flow to infinitesimal 3D structural perturbations (e.g., Tollmien-Schlichting waves), triggering a transition to turbulence that invalidates the 1D simplified kinematic profile.

14.5 Laminar Shear Flow Between Parallel Plates

Consider the steady flow of a viscous fluid between two infinite parallel plates separated by a distance of $2h$ (located at $y = -h$ and $y = h$).

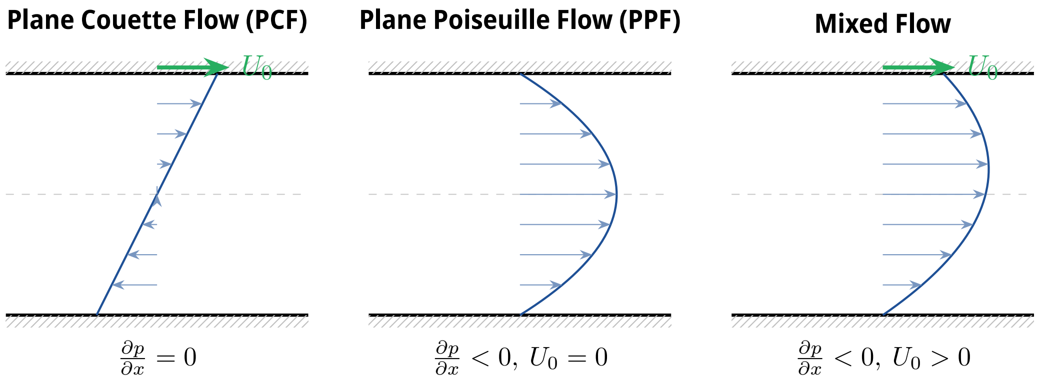


Figure 2: Three standard regimes of steady laminar flow between parallel plates.

We assume a steady, parallel, laminar flow with the velocity vector oriented entirely in the downstream direction:

$$\vec{v}(x, y, z, t) = u(y, t)\hat{i} \tag{174}$$

By applying this velocity profile to the mass conservation equation:

$$\nabla \cdot \vec{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \implies \frac{\partial u}{\partial x} = 0 \tag{175}$$

which is identically satisfied since u does not depend on x.

Next, we evaluate the convective term:

$$\vec{v} \cdot \nabla \vec{v} = \left(u \frac{\partial}{\partial x} \right) u(y, t)\hat{i} = \vec{0} \tag{176}$$

Thus, the non-linear advection term vanishes identically.

Substituting this into the components of the Navier-Stokes equations:

$$x\text{-momentum:} \quad \rho \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} \tag{177}$$

$$y\text{-momentum:} \quad 0 = -\frac{\partial p}{\partial y} - \rho g \implies p(x, y, t) = -\rho g y + f(x, t) \tag{178}$$

$$z\text{-momentum:} \quad 0 = -\frac{\partial p}{\partial z} \tag{179}$$

Since the left-hand side of Eq. (177) is independent of x, and $\mu \frac{\partial^2 u}{\partial y^2}$ is independent of x, the pressure gradient $\frac{\partial p}{\partial x}$ must be independent of x. Thus, we can write the integrated pressure field as:

$$p(x, y, t) = -\rho g y + \left(\frac{\partial p}{\partial x} \right) x + C \tag{180}$$

This leaves the governing momentum equation as a classic one-dimensional diffusion equation:

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} - \frac{1}{\rho} \frac{\partial p}{\partial x} \tag{181}$$

where $\nu = \mu/\rho$ is the kinematic viscosity, representing the momentum diffusion coefficient.

For a steady-state flow ($\frac{\partial u}{\partial t} = 0$), the governing equation simplifies to:

$$\mu \frac{d^2 u}{dy^2} = \frac{\partial p}{\partial x} \tag{182}$$

Integrating twice with respect to y yields:

$$u(y) = \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) y^2 + C_1 y + C_2 \tag{183}$$

Applying the boundary conditions $u(-h) = 0$ (stationary bottom plate) and $u(h) = U_0$ (top plate moving with speed U_0):

$$u(h) = \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) h^2 + C_1 h + C_2 = U_0 \tag{184}$$

$$u(-h) = \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) h^2 - C_1 h + C_2 = 0 \tag{185}$$

Solving this linear system for the constants C_1 and C_2:

$$C_1 = \frac{U_0}{2h}, \quad C_2 = \frac{U_0}{2} - \frac{h^2}{2\mu} \left(\frac{\partial p}{\partial x} \right) \tag{186}$$

General Velocity Profile for Mixed Plate Flow

$$u(y) = \underbrace{\frac{U_0}{2} \left(1 + \frac{y}{h} \right)}_{\text{Linear (Couette)}} + \underbrace{\frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) (y^2 - h^2)}_{\text{Parabolic (Poiseuille)}} \tag{187}$$

The volumetric flow rate per unit depth Q is computed by integrating the velocity profile across the gap:

$$Q = \int_{-h}^{h} u(y)\, dy = \int_{-h}^{h} \left[\frac{U_0}{2} \left(1 + \frac{y}{h} \right) + \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) (y^2 - h^2) \right] dy \tag{188}$$

Evaluating the integrals:

$$\int_{-h}^{h} \frac{U_0}{2} \left(1 + \frac{y}{h} \right) dy = U_0 h \tag{189}$$

$$\int_{-h}^{h} \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) (y^2 - h^2)\, dy = \frac{1}{2\mu} \left(\frac{\partial p}{\partial x} \right) \left[\frac{y^3}{3} - h^2 y \right]_{-h}^{h} = -\frac{2h^3}{3\mu} \left(\frac{\partial p}{\partial x} \right) \tag{190}$$

Result: Volumetric Flow Rate

$$Q = U_0 h - \frac{2h^3}{3\mu} \left(\frac{\partial p}{\partial x} \right) \tag{191}$$

14.6 Hagen-Poiseuille Pipe Flow

Next, we analyze steady, laminar flow through a straight pipe of circular cross-section with radius R.

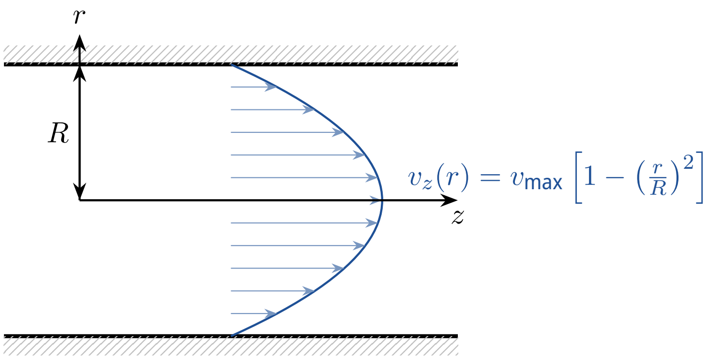


Figure 3: Parabolic velocity profile of Hagen-Poiseuille flow in a circular pipe.

We employ cylindrical coordinates (r, θ, z) and assume an incompressible, steady, axisymmetric, parallel flow:

$$\vec{v} = v_z(r)\hat{e}_z \tag{192}$$

By applying this velocity field to the cylindrical continuity equation:

$$\nabla \cdot \vec{v} = \frac{1}{r}\frac{\partial}{\partial r}(r v_r) + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0 \implies \frac{\partial v_z}{\partial z} = 0 \tag{193}$$

which is identically satisfied. Under these assumptions, the z-momentum equation simplifies to:

$$0 = -\frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right)\right] \tag{194}$$

Since the pressure gradient is constant along the pipe length, we integrate the equation directly:

$$\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = \frac{r}{\mu}\left(\frac{\partial p}{\partial z}\right) \tag{195}$$

Integrating once with respect to r:

$$r\frac{dv_z}{dr} = \frac{r^2}{2\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \implies \frac{dv_z}{dr} = \frac{r}{2\mu}\left(\frac{\partial p}{\partial z}\right) + \frac{C_1}{r} \tag{196}$$

Integrating a second time yields:

$$v_z(r) = \frac{r^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \ln r + C_2 \tag{197}$$

To determine the integration constants, we apply physical boundary conditions:

- **Centerline Regularity:** The velocity must remain finite at the pipe centerline ($r \to 0$). Since $\lim_{r\to 0} \ln r = -\infty$, we must set:

$$C_1 = 0 \tag{198}$$

- **No-Slip Condition at the Wall ($r = R$):** $v_z(R) = 0$

$$\frac{R^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_2 = 0 \implies C_2 = -\frac{R^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) \tag{199}$$

Hagen-Poiseuille Velocity Profile

$$v_z(r) = -\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)(R^2 - r^2) \tag{200}$$

The volumetric flow rate Q is computed by integrating this profile across the circular cross-section:

$$Q = \int_A v_z\, dA = \int_0^{2\pi}\int_0^R v_z(r)\, r\, dr\, d\theta = 2\pi \int_0^R v_z(r)\, r\, dr \tag{201}$$

Substituting Eq. (14.6):

$$Q = -\frac{\pi}{2\mu}\left(\frac{\partial p}{\partial z}\right)\int_0^R (R^2 r - r^3)\, dr = -\frac{\pi}{2\mu}\left(\frac{\partial p}{\partial z}\right)\left[\frac{R^2 r^2}{2} - \frac{r^4}{4}\right]_0^R \tag{202}$$

Result: Hagen-Poiseuille Law

The volumetric flow rate is proportional to the pressure gradient and scales with the fourth power of the pipe radius:

$$Q = -\frac{\pi R^4}{8\mu}\left(\frac{\partial p}{\partial z}\right) \tag{203}$$

14.7 Annular Flow

We extend this analysis to viscous flow through an annular channel bounded by an inner cylinder of radius R_i and a coaxial outer cylinder of radius R_o.

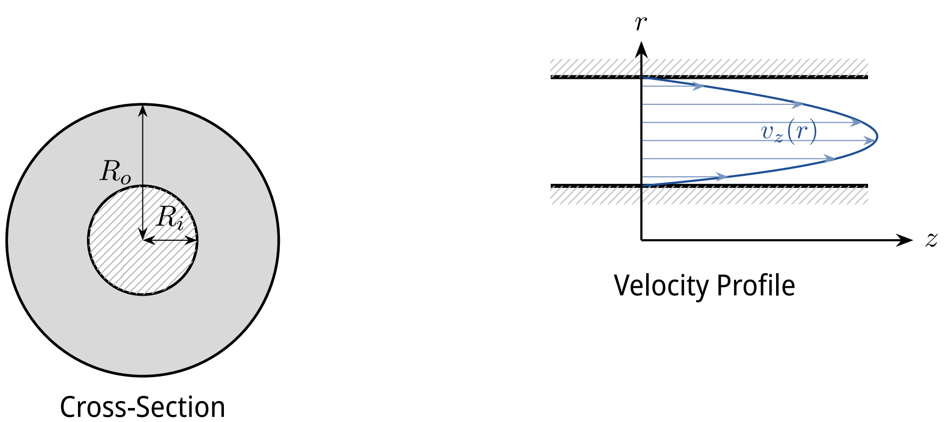


Figure 4: Geometry and velocity profile for laminar flow in a coaxial annulus.

The general solution of the z-momentum equation remains:

$$v_z(r) = \frac{r^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \ln r + C_2 \tag{204}$$

However, because the domain does not include the origin $r = 0$, the logarithmic term is physically valid and $C_1 \neq 0$. We apply the no-slip condition at both solid boundaries:

$$v_z(R_i) = 0 \implies \frac{R_i^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \ln R_i + C_2 = 0 \tag{205}$$

$$v_z(R_o) = 0 \implies \frac{R_o^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \ln R_o + C_2 = 0 \tag{206}$$

Subtracting Eq. (205) from Eq. (206) to solve for C_1:

$$\frac{R_o^2 - R_i^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + C_1 \ln\left(\frac{R_o}{R_i}\right) = 0 \implies C_1 = -\frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)\frac{R_o^2 - R_i^2}{\ln(R_o/R_i)} \tag{207}$$

Substituting C_1 back into Eq. (206) to solve for C_2:

$$C_2 = -\frac{R_o^2}{4\mu}\left(\frac{\partial p}{\partial z}\right) + \frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)\frac{R_o^2 - R_i^2}{\ln(R_o/R_i)} \ln R_o \tag{208}$$

Recombining terms into the general velocity equation:

$$v_z(r) = \frac{1}{4\mu}\left(\frac{\partial p}{\partial z}\right)\left[r^2 - R_o^2 - \frac{R_o^2 - R_i^2}{\ln(R_o/R_i)} \ln r + \frac{R_o^2 - R_i^2}{\ln(R_o/R_i)} \ln R_o\right] \tag{209}$$

Velocity Profile for Annular Flow

$$v_z(r) = \frac{1}{4\mu}\left(-\frac{\partial p}{\partial z}\right)\left[R_o^2 - r^2 - \frac{R_o^2 - R_i^2}{\ln(R_o/R_i)} \ln\left(\frac{R_o}{r}\right)\right] \tag{210}$$

Expert Analysis: Asymmetric Profile Analysis

Unlike simple pipe flow, the velocity profile in an annulus is asymmetric. The position of maximum velocity r_{max} is shifted towards the inner wall due to the higher viscous shear area near the core. Setting $\frac{dv_z}{dr} = 0$:

$$-2r_{\text{max}} + \frac{R_o^2 - R_i^2}{\ln(R_o/R_i)}\frac{1}{r_{\text{max}}} = 0 \implies r_{\text{max}} = \sqrt{\frac{R_o^2 - R_i^2}{2\ln(R_o/R_i)}} \tag{211}$$

This mathematically confirms that the location of peak velocity is closer to the inner boundary than the arithmetic mean radius of the channel.